

An Event-Dated Corpus of US Trademark Prosecution and a Two-Sided Measure of Vocabulary Position

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Abstract

The USPTO publishes every US trademark application since 1884 with its prosecution history. We re-parse all 13.99 million case files into 242 million dated events, separating maintenance deadlines that terminal status codes conflate. Dating each cancellation makes the sworn five-year proof of continued use a product-level survival outcome: whether a registered mark’s goods were still sold. We then build a two-sided measure of vocabulary position: a topic model reduces each goods/services description to themes, and each theme mix is compared by Kullback–Leibler divergence with its own class’s filings in the five years before and after. Their average is *atypicality*, how unusual the language is; their difference is *lead*, whether it ran ahead of or behind the industry. Rescored under three vocabulary partitions, two reference weightings, five window constructions, a refitted model, and duplicate-text removal, lead survives every variation and predicts cancellation at the five-year proof (+1.4 points with design controls, $t = 8.3$); atypicality’s effects belong to the partition or to a control. Two hazards follow: scoring words rather than themes measures description length ($r = -0.64$), and firm-level correlations built on word scoring dissolve under theme scoring. Corpus, scores and code are public.

Keywords: trademarks; innovation indicators; text as data; topic models; administrative data; product survival.

1 Introduction

A US trademark application carries a description of the goods or services the mark will cover. That description is written to secure scope of protection, it is dated, it is public, and it is filed by the millions of firms that never patent. The literature that reads trademarks as innovation data has worked mostly from what surrounds the description: how many marks a firm files, in which classes, how broadly, and whether it renews them (Castaldi, 2020; Dinlersoz et al., 2018; Flikkema et al., 2019). The text itself has entered only piecemeal, as breadth counts or similarity scores. This paper builds two research assets from the description and the prosecution record around it, states what each can and cannot support, and makes both public.

The first asset is a corpus. The USPTO publishes every application since 1884 with its full prosecution history, but the literature has mostly read one field from it: the terminal status code. Status codes run together events that the maintenance schedule keeps apart. Between the fifth and sixth year after registration, an owner must file a sworn declaration that the mark is still in use in commerce, and must renew again at year ten; a registration whose owner does neither is cancelled, and both cancellations share a status code. We re-parse all 13.99 million case files into 242 million dated events, so each cancellation carries its date and the record shows which deadline

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the mark failed. The five-year proof of continued use thereby becomes a dated, sworn, product-level survival record, available for every registered mark more than seven years old. The same parse records office actions and responses, counsel of record, filing basis and owner domicile, and owners are linked by normalized name to SEC reporting, Regulation D rounds, and PatentsView patent assignees. Where the existing public datasets supply case-file snapshots (Graham et al., 2013) or Census-linked registration histories (Dinlersoz et al., 2018), the event-dated corpus adds the full prosecution timeline of each case, and with it the dated outcomes.

The second asset is a measure. A topic model, fitted once on a sample drawn equally from every industry and then applied to every filing, reduces each description to a mix of themes — whether the filing describes an online offering, say, or an artificial-intelligence application — and we compare each filing’s theme mix, by Kullback–Leibler divergence, with what its own Nice class — the international classification’s 45 classes of goods and services — filed in the five years before its date and in the five years after. Their average is the filing’s *atypicality*: how unusual its language is for its industry. Their difference is its *lead*: whether its language sat ahead of or behind where the industry’s language was moving. Looking both backward and forward is what separates *unusual* from *early*, and the two components predict different things.

We validate the measure by variation rather than by a single fit. The identical filings are rescored under three partitions of the vocabulary, two reference weightings, five window constructions, a refit of the theme model under a second random seed (in the software class), and with and without the 47% of registrations whose text is shared verbatim with another filing. Lead survives every variation and predicts failure at the five-year proof: +1.4 points from the most lagging to the most leading fifth with class-by-cohort fixed effects, drafting and filer controls and owner-clustered errors ($t = 8.3$); +2.3, +2.6 and +1.9 points raw under the three partitions. Atypicality does not survive: its apparently protective effect at the proof falls from 4.4 points under global themes to 0.7 under per-class themes, and its association with reaching SEC reporting vanishes once description length is held.

Two hazards follow for any research that scores trademark text, and we document both on identical samples. Scoring words rather than theme distributions measures description length: term-scored atypicality correlates with log distinct terms at -0.64 within class. And firm-level correlations built on word scoring — the kind the literature reports — dissolve or reverse under theme scoring on the same firms; what they carry is unsigned lexical specificity, not vocabulary position.

What the measure predicts about pioneering and industry waves is the subject of a companion paper; here those outcomes appear only as validation, and the paper makes no claim about mechanism. Section 2 describes the corpus and the outcomes it dates. Section 3 builds the measure. Section 4 settles nomenclature against the innovation literature’s terms. Section 5 tests the measure against registration, the five-year proof, patenting and SEC reporting. Section 6 summarizes what changes under every alternative construction tried, and Section 7 states the practices that follow for trademark-text research.

2 The record: from case files to dated events

2.1 Corpus and parse

The USPTO Trademark Annual Applications backfile (product TRTYRAP; United States Patent and Trademark Office, 2026) is the complete record of US trademark filings from 1884 onward: 13.99 million case files, each carrying the goods/services description by class, the prosecution history, and the ownership and correspondence records. The prosecution history is a list of coded events with

dates. We re-parse every case file into 242 million dated events, keyed by serial number, so that each application’s full sequence is available: office actions, responses, allowances, statements of use, registration, Section 8 and Section 9 filings, cancellations and renewals, each with its date.¹ The parse also reads counsel of record on each filing, the statutory filing basis (use, intent to use, §44(e) home registration, §66(a) Madrid extension), the owner’s domicile, and the owner name, which is normalized — uppercased, stripped of punctuation and of two dozen legal suffixes — so that ACME CORP. and ACME CORPORATION are one owner. Graham et al. (2013) describe the administrative data; the event-dated parse adds the full prosecution timeline of each case.

2.2 The steps, the samples, and the conventions

A US trademark passes through a sequence of dated, high-stakes steps. An application is filed with a goods/services description. An examiner reviews it, often issuing office actions that the applicant must answer. If the examiner allows the application — and, for intent-to-use filings, once the applicant files a sworn statement that the mark has entered commerce — the mark registers. Registration then starts a maintenance clock. In years five to six the owner must file a Section 8 declaration swearing that the mark is still in use in commerce, with a six-month grace period; this is the *five-year proof of continued use*, the paper’s central outcome. Around year ten a combined Section 8 declaration and Section 9 renewal is due, and again every ten years after that. A mark whose owner misses a proof is cancelled. Madrid Protocol registrations (§66(a) basis) file Section 71 declarations on the same schedule and are flagged separately throughout.

Table 1: The trademark funnel in this corpus. Registration counts are unique serials; five-year-proof outcomes are event-dated (a Section 8 cancellation at registration age 4.0–8.5 years). Source: the re-parsed USPTO application backfile.

Stage	<i>n</i>	Share
Debut filings scored, 1995–2018 (owners’ first filings)	3,589,068	
reached registration	2,066,600	57.6%
All unregistered class-records filed 1995–2018	3,798,318	
never filed statement of use		42.6%
stopped responding in prosecution		45.3%
removed adversarially		4.3%
Registrations 2002–2018, scored, unique serials	3,104,534	
failed the five-year proof of continued use (event-dated)		52.3%
Passed the five-year proof, cohorts 2002–2013, status known	1,129,724	
failed at the year-ten renewal		39.1%

Three named outcomes recur (Table 1). *Failing to register*: the application never completed, which happens by the applicant’s own inaction in roughly nine cases of ten. *Failing the five-year proof of continued use*: a registration cancelled for non-use at registration age 4.0 to 8.5 years, which means the product the mark named left commerce within six years. *Reaching the SEC record*: the owner appears in dated SEC events.

¹Event codes are read against the USPTO’s own event-code dictionary. Terminal status codes are retained for cross-checking: 700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired. The year-six cancellation and the year-ten cancellation share status 710 and are separable only by event date, which is the reason the corpus is event-dated.

Outcomes are rates, so every estimate is a shift in a probability and is stated in percentage points against the base rate it shifts. The basic comparison is the outcome rate of the fifth of filings with the highest lead minus that of the fifth with the lowest, with the fifths cut inside a single Nice class and year so that no difference between industries or between years enters; regression tables add drafting and filer controls and cluster errors on the owner. A filing may claim several Nice classes at once, so a *class-record* counts it once per class while a *registration* counts it once by serial number; a *debut filing* is an owner’s first; a *cohort* is the set of registrations issued in a calendar year.

Two windows bound the samples, each set by an institutional fact. Scoring runs 1995–2019, because a filing can only be scored when the record holds a full five years of its class’s language on either side of its date. Full windows exist for filings made 1990 through 2020; scoring holds back five years at the start, where references are thinnest, and one at the end. The supplementary material (§S3) shows how thin references distort scores near the start of the record. Five-year-proof cohorts run 2002–2018: cancellations post at about registration age six and a half, and the record ends in April 2026, so 2018 is the last cohort old enough for its proof window to have essentially elapsed.

2.3 Owner links

Owners are matched to three external records by normalized exact name. *SEC reporting* is any appearance of the owner as a registrant in the Commission’s EDGAR filer index (U.S. Securities and Exchange Commission, 2026a,b); roughly half the matched firms carry a ticker and the rest report for other reasons. *Regulation D* rounds are Form D notices of exempt offerings (U.S. Securities and Exchange Commission, 2026c), electronic and therefore complete only from March 2009. *Patents* are PatentsView assignee records (PatentsView, 2024). The SEC match runs over all 5.67 million owners in the corpus; names resolving to more than one registrant are dropped rather than assigned, yielding 19,889 matched owners covering 7,588 registrants. Validation on 22 companies that indisputably went public matches all 22, and a hand inspection of 25 random matches found none wrong — a check on gross failure modes rather than a precision estimate. The linkage is nonetheless a name match, and that bounds what can be read from it: distinctive names are easier to match, and distinctiveness of a name plausibly travels with distinctiveness of the text. Section 5 returns to which results survive that bound.

3 The measure

3.1 Two readers, one living backwards

Imagine two readers of a filing’s goods/services description, each having seen a disjoint half of its industry. One has read only what the industry filed in the five years before it. The other, living backwards as Merlin was said to, has read only the five years after. Each is asked how surprising the wording is. The measurement is how much they disagree: a description that startles the first reader and not the second arrived before its industry’s language moved that way.

Both windows are anchored on the filing’s own date, spanning the 1,826 days before it and the 1,826 days after, and both exclude the filing date itself. Each window holds every application filed in the same Nice class in that span, registered or not, and every one counts the same; a filing is scored only when each of its windows holds at least 500 filings. The measure asks whether wording is unusual against a period, not whether it is ahead of a trend inside that period; it has no momentum channel.

3.2 What is measured

The quantity is information content in Shannon’s sense. Something that happens with probability p carries $\log(1/p)$ of information when it happens: a common thing carries little, a rare thing a lot. Here the things are the themes a description draws on, and the probabilities are how often the filing’s own industry drew on each of them in the reference window. A filing’s score asks, in effect: how surprising is this mix of themes to someone who has only seen what the industry files?

The apparatus is that of Murdock et al. (2017), developed on Darwin’s reading notebooks, and Barron et al. (2018), who extended it to French Revolution parliamentary debates; they call the two quantities *novelty* and *transience* and their signed difference *resonance*. The arithmetic carries over unchanged; the interpretation does not, and Section 4 renames the quantities for a commercial record in which the author is usually counsel. One usage note carries through everywhere: in those settings latent Dirichlet allocation identifies *topics* — what a document is about. A goods/services description is not about anything in that sense; it lists components, and across millions of filers a cluster of co-occurring terms (*video, music, games, electronic, audio*) is a component many offerings share. *Theme* is the word used throughout; *topic* appears only in the model’s name.

3.3 From a description to two numbers

Counting words. A description is reduced to a bag of words: lowercased, split into tokens of three or more letters, and counted, with adjacent pairs counted as tokens in their own right so that *computer software* registers alongside *computer* and *software*. Take the filing for AMAZON.COM, lodged in the advertising and retail class on 18 April 1997:

computerized on line search and ordering service featuring the wholesale and retail distribution of books, music, motion pictures, multimedia products and computer software in the form of printed books, audiocassettes, videocassettes, compact disks, floppy disks, CD ROMs, and direct digital transmission

Of its 65 distinct words and word pairs, 35 are in the theme model’s vocabulary, among them *computerized, search, ordering service, retail* and *digital transmission*.

Grouping words into themes. Next, each description is re-described using a short list of themes instead of tens of thousands of separate words. A theme here is simply a list of words that tend to turn up in the same descriptions: *video, computer, electronic, music, games*, say, or *store, retail, store services, featuring*. We do not write these lists ourselves. They are found by a standard statistical method, latent Dirichlet allocation (Blei et al., 2003), which works roughly as follows. It starts from a random guess about which words belong to which theme, then passes over the descriptions again and again, each time adjusting the word lists so that two things become true together: each description can be explained by a few themes, and each theme uses a consistent set of words. When the lists stop changing, each theme is a weighted list of words, and each description is a mix of themes — for example 40% of one, 15% of another — that sums to one. Nothing tells the model what the themes should be about; we name them afterwards by reading their leading words. The online appendix lists every theme with its leading words and its weight in each class.²

Three choices go into the model, and each affects what it can see. *What it learns from.* The model is fitted once, on a sample of 10,000 descriptions from each of the 45 Nice classes (every description, for classes smaller than that), 448,437 in all, and then applied to all 13.99 million. Sampling each class equally keeps the largest classes — software and electronics alone holds 1.8

²https://aporia.institute/tm-vocabulary/online-appendix/themes_T50.html

million filings — from dominating the theme list, so that small industries get themes of their own. The cost is that vocabulary concentrated in the largest classes is thinned before the model sees it. Descriptions using internet terms are 7.4% of the sample against 14.7% of the corpus; for cloud computing the figures are 0.4% against 2.4%, for artificial intelligence 0.4% against 0.7%, and for blockchain 0.2% against 0.5%, while solar vocabulary, spread over many small classes, enters at its true rate. Neither the 50- nor the 500-theme model has a dedicated theme for artificial intelligence, blockchain or cloud computing, which is why analyses of those waves use curated word lists rather than themes. *Which words count.* The vocabulary is every word, and every pair of adjacent words, that appears in at least 50 sampled descriptions, after common function words such as *the* and *for* are removed: 62,168 terms.³ *How many themes.* The number is set by the analyst, and we use fifty.

Fifty is coarse — roughly one theme per Nice class — and it is chosen on three grounds. First, at fifty every theme is a coherent word list. A standard check is whether a theme’s ten leading words actually appear in the same descriptions more often than chance would put them there (normalized pointwise mutual information, which runs from -1 to 1 , with 0 meaning no association; Lau et al., 2014). At fifty themes the average is 0.48 and no theme falls below zero. At 200 themes the average falls to 0.33 and 11% of themes fall below zero; at 500 themes, to 0.23 and 21% — a fifth of the finer model’s themes are word lists whose words rarely occur together. Finer models fit the text more closely, but a closer statistical fit does not by itself make themes more interpretable (Chang et al., 2009). Second, the themes themselves come back when the model is refitted. Refitting with a different random seed and matching each theme to its closest counterpart, the median theme’s word list has a cosine similarity of 0.74 with its match at fifty themes, against 0.54 at 200. The per-filing scores are a noisier matter at any resolution: in the software class a second seed reproduces lead at $r = 0.72$ at fifty themes (0.79 at 200), and moves the five-year-proof contrast from $+3.2$ to $+3.9$ points. Each fit is therefore a noisy measurement of lead, and noise of this kind pulls estimates toward zero rather than creating them. Third, and decisively, the findings do not depend on the choice. Section 5.2.1 rescores every registration under 500 global themes and under 50 themes fitted separately inside each class — 2,250 class-specific themes in all, the finest partition tested — and the headline contrast is $+2.3$, $+2.6$ and $+1.9$ points under the three, and $+2.2$ at 200 themes (Supplementary Material, §S2 and §S4). Fifty is the resolution at which the themes are easiest to read and every result holds; nothing in the paper rests on the model having found the one right number.

This build — fifty global themes, per-filing five-year windows, flat weighting — is the production scoring, used for every estimate unless the text states another. The themes are shared across classes; what is specific to a class is the reference, the average theme mix of that class’s own filings in the window. The Amazon filing draws on six themes at more than a twentieth of its weight: 0.41 on media and entertainment goods, 0.15 on computer and information services, 0.13 on retail store services, 0.10 on printed matter, 0.09 on downloadable and online software, and 0.07 on marketing of consumer goods.

³The fit uses scikit-learn’s online variational implementation with its default priors, eight passes over the sample, and a fixed random seed; the sample is drawn from filings dated 1990–2024. When filings are scored, each description is read with a simpler tokenizer than the one used for fitting (words of three or more letters, function words kept, word pairs formed across punctuation), so a word pair the model learned across a removed function word never occurs at scoring time. Every filing is scored by the same rule, so the difference is a fixed feature of the measure rather than a source of bias between filings. On a sample of 37,124 descriptions from every class, the two tokenizations give the same dominant theme for 88% of descriptions and theme mixes that differ by a median of 0.07 in total variation (on a 0 – 1 scale).

Comparing against the two references. Write P for the filing’s distribution over themes, and Q^- and Q^+ for that distribution averaged over every same-class filing in the two windows. How far P sits from a reference Q is the Kullback–Leibler divergence,

$$\text{KL}(P \parallel Q) = \sum_k P_k \log \frac{P_k}{Q_k},$$

zero when the two distributions agree and growing as the filing puts weight where the industry puts little, measured in nats. Write $K^- = \text{KL}(P \parallel Q^-)$ and $K^+ = \text{KL}(P \parallel Q^+)$: the filing’s past-facing and future-facing surprise, what the first reader feels and the second.

Taking the difference. Amazon’s filing is scored against 41,166 earlier filings and 146,532 later ones, giving $K^- = 1.286$ and $K^+ = 1.163$. The two axes used throughout are their average and their signed difference:

$$\underbrace{A = \frac{1}{2} (K^- + K^+)}_{\text{atypicality}}, \quad \underbrace{L = K^- - K^+}_{\text{lead}}.$$

For Amazon, $A = 1.225$ and $L = +0.123$: against its own class, its atypicality is at the 47th percentile and its lead at the 94th. The industry moved toward this description in the five years after it was filed and had not been writing that way in the five years before.

Why the average and the difference, rather than the two surprises themselves? Because K^- and K^+ correlate at 0.99: a filing far from its class’s past is almost always far from its future too, so entering both is close to entering one variable twice. The average collects that large shared component and the difference collects the small residual that carries the timing; nothing more than a sum and a difference is taken, with no weights estimated. The two resulting axes are only weakly related: within class, lead and atypicality correlate at -0.10 across all scored filings, though the figure varies by class, from $+0.04$ in software to $+0.25$ in advertising and retail, and Section 5.2.1 checks that the lead results do not run through atypicality. The cost of the construction is that L is a small residual of two large quantities, with about a seventh of either level’s spread, which the supplementary material (§S4) quantifies.

Why the words are not scored directly. The words themselves could be compared to the reference, and the paper reports that scoring as a cross-check. It cannot be the primary one, because a divergence measured on words mostly counts the length of the filing. For any two distributions,

$$\text{KL}(P \parallel Q) = \underbrace{-\sum_k P_k \log Q_k}_{\text{how unexpected these words are}} - \underbrace{(-\sum_k P_k \log P_k)}_{\text{how spread out the filing is}},$$

and the spread of a filing over words is capped by how many distinct words it uses: a description with k distinct terms cannot spread past $\log k$, while its class’s reference spreads over thousands. A short description is therefore charged for its narrowness before any of its content is considered, and the shorter the description, the larger the charge. In the data, term-scored atypicality correlates with the log of a filing’s distinct-term count at -0.65 in the software class and -0.64 across all classes (within class). Themes break the coupling because all 50 coordinates are always occupied: theme-scored atypicality correlates with the same count at between -0.10 and $+0.10$. What words can show and themes cannot is specific new wording — a word pair like Amazon’s *line search* has no place in any theme — so theme scoring carries every estimate, and the worked examples in Supplementary Material §S2 trace three filings through both representations.

What the themes see, and what they do not. The score reads which themes a filing draws on and how heavily, not how it combines them: on a 150,000-filing software-class sample, the rarity of a filing’s two leading themes explains 70% of the variance in atypicality and the unusualness of the pairing adds under two points. A filing assembled from ordinary parts in a genuinely new configuration is close to invisible to this measure. Scoring arrangements directly — a pair-level lead, and word-level recombination in the sense of Fleming (2001); Uzzi et al. (2013) — shows the two channels acting in opposite directions in the software class and neither generalizing corpus-wide, so the construct’s blindness to combination is a bounded omission rather than a hidden bias in one direction (full estimates in Supplementary Material §S2).

3.4 Within class, not across it

Goods/services descriptions are drafted to secure scope of protection, and counsel reuse language from the USPTO’s own Acceptable Identification Manual, from house precedent, and from competitors’ successful filings. Identical wording across unrelated firms is therefore common: 17% of scored registrations share a value with another filing in their class and registration year. Within a class, the measure picks up departure from a shared drafting convention. Across classes it does not: a class where the Manual supplies dense standard language shows a compressed distribution of atypicality, and one where firms describe novel services in prose a wide one, for reasons unrelated to how innovative either industry is. Every estimate is therefore taken within class and year, and magnitudes compare across classes only as signs and orderings. The vocabulary is positional rather than causal: a leading filing was ahead of its industry’s language, which implies a trajectory without asserting the filing caused it.

4 Neighbours and nomenclature

4.1 Trademarks as innovation data, and text measures of novelty

A trademark filing is a dated, costly, public disclosure made by the millions of firms that never patent, and a literature has grown around reading it as an innovation indicator (Castaldi, 2020; Willeke et al., 2026). Trademark counts and breadth predict venture valuations (Block et al., 2014); filing timing tracks innovation processes (Seip et al., 2018); trademarks mark the product-market side of innovation that patents miss (Flikkema et al., 2019, 2025); and Census-linked work reads registration as a correlate of firm growth and survival (Dinlersoz et al., 2018, 2023). Nearly all of it counts filings, classes, or renewals. This paper reads the text itself, and reads it against time.

Scoring a document by its divergence from a reference corpus before and after it comes from cultural analytics (Barron et al., 2018; Murdock et al., 2017). In economics and finance, text-based measures locate firms in product space (Hoberg & Phillips, 2016), date technological innovation in patents over the long run (Kelly et al., 2021), and track the vintage of patent language (Kalyani, 2024). The construction here differs in its object — the composition of commercial offerings, not ideas or claims — and in its two-sided reference: every filing is scored against its own industry’s language both before and after its date, which is what lets *unusual* be separated from *early*.

4.2 Nomenclature

The innovation literature is not consistent about what quantities like these are called, and several of its established terms assert things this design cannot show, so the paper’s names are chosen against the nearest occupants rather than borrowed from them (Table 2).

Table 2: The five quantities, their nearest occupants in the literature, and the terms used here. P is a filing’s theme distribution; the references Q^- and Q^+ aggregate the five years of same-class filings before and after it. *Lagging* and *leading* describe the two signs of L . The level names appear only where the two-sided construction itself is at issue; the analysis runs on A and L .

Quantity	Nearest in the literature	Term used here
$K^- = \text{KL}(P Q^-)$	novelty (Barron et al., 2018; Murdock et al., 2017); originality (Hall et al., 2001)	past-facing surprise
$K^+ = \text{KL}(P Q^+)$	transience (Murdock et al., 2017); generality (Hall et al., 2001), which measures realized influence, not position	future-facing surprise
$A = \frac{1}{2}(K^- + K^+)$	atypicality (Uzzi et al., 2013), which scores internal pairings, not distance from the class	atypicality
$L = K^- - K^+$	resonance (Barron et al., 2018), which asserts a mechanism	lead (leading / lagging)
$ L $	no established occupant	lead magnitude

Novelty is declined as a name for anything here because it names only the backward-looking half of a quantity that faces both ways. *Resonance* is arithmetically what this paper calls lead, and is avoided because it names a mechanism — the field resonating with an author’s contribution — that this design cannot observe: a filing can sit where its class was heading because it was imitated, because both responded to the same outside event, or by coincidence. *Radicalness* is declined because it claims existing competences are devalued, which nothing here licenses, and because the label itself is unsettled: a bibliometric review of 4,407 articles finds *radical*, *breakthrough*, *discontinuous* and *disruptive* used with definitions inconsistent within labels and not distinctive across them (Verhoeven et al., 2025). *Atypicality* in the bibliometric sense (Berkes & Gaetani, 2021; Uzzi et al., 2013) scores the rarity of the pairwise combinations inside a document; the atypicality here is the distance of one document’s composition from its class’s, and it barely registers how unusual a filing’s internal pairings are (Section 3). The word is kept because both mean unlike the population; which population, and along what, differ. *Generality* (Hall et al., 2001) is the nearest established forward-facing quantity to K^+ , but it requires forward citations to have arrived, so it observes realized influence; K^+ observes the class’s movement whether or not the filing influenced it. *Distinctiveness* is occupied twice, by trademark law’s registrability doctrine and by optimal distinctiveness in strategy (Zhao et al., 2017), whose predicted inverted-U appears here at registration, so the term would imply a theory is being tested when it is not. What *atypicality* says is narrower than any of these: this description is unusual for its class and year. Whether the thing described was new, radical, or good is not measured.

5 Validation against the record’s outcomes

A measure is worth the outcomes it can be tested against. The record supplies four that are dated and high-stakes: whether an application reached registration, whether a registered mark’s owner proved continued use at year six, whether the owner patents, and whether the owner appears in SEC reporting. This section reports what each axis predicts at each, under the design the corpus supports — class-by-year fixed effects, drafting and filer controls, owner-clustered errors — and

states, for each result, whether it survives the alternative constructions of Section 6. What the results say about pioneering is the subject of the companion paper.

5.1 Registration rises with atypicality and is nearly flat in lead

Figure 1 gives the registration rate by decile of each axis, deciles cut within Nice class and filing year. On atypicality the rate rises from 56.0% in the least atypical decile to 59.1% in the ninth and turns down to 58.3% in the most atypical. On lead the range across deciles is 0.9 points: filings written in language their class was moving toward register at nearly the same rate as filings written in language it was leaving, though a linear fit tilts slightly upward.

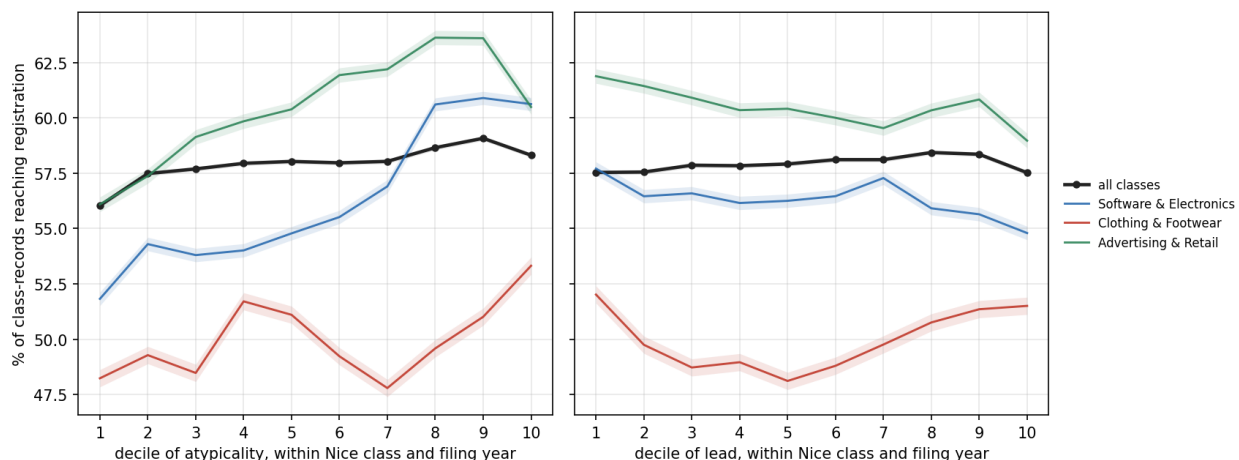


Figure 1: Share of class-records reaching registration, filings 1995–2018, by decile of atypicality (left) and of lead (right), deciles cut within Nice class and filing year; $n = 9,024,532$ scored class-records. The black line pools all 45 classes; the colored lines are three named classes. The grey band is a 95% binomial interval on the pooled line.

The flatness of lead at this step has a mechanical reading worth stating: lead is computed from the five years of filings *after* the filing date, which had not happened when the examiner acted. No participant in prosecution could observe the quantity. Atypicality’s past-facing half, by contrast, is exactly what an examiner steeped in the class’s usual language experiences, and it is the axis registration responds to.

Two cautions on the level. First, registration is mostly a measure of the applicant’s own follow-through: nine in ten failed applications die by inaction, and only 4.3% are removed adversarially (Table 1). Second, the shape depends on the cut. On the pooled distribution, restricted to debut filings, the atypicality curve is an inverse-U with a deeper conventional tail (46.6% at the bottom, about 58% in the middle, 53.4% at the top); the within-cell profile of Figure 1 is the cleaner statement, and on it the conventional tail, not the atypical one, is the penalized end. Both shapes reward a middling distance from the class’s language — the pattern audience-conformity research predicts for evaluators (Hsu, 2006; Zhao et al., 2017; Zuckerman, 1999), here from an examiner at census scale. Unfamiliarity alone cannot produce it, since completion falls at both ends, and the curvature is specific to the unsigned axis (Supplementary Material, §S1, carries the debut profiles, the counsel-vs-self-filed split, and refileing behaviour).

5.2 The five-year proof: leading marks fail it more often

The question here is about products that entered commerce: given that a product was brought to market, does the language of its first description predict whether it was still there five years on? Registration is the record’s marker of entry, so the analysis conditions on it. An application that never registered left the record, nine times in ten, by the applicant’s own inaction, before any product met a customer; pooling those applications in would change the question into one about everyone who considered launching. Refiling does not undermine the conditioning: 3.6% of abandoned applications are refiled within six years, refiled cases are 1.1% of this sample and fail at nearly the sample rate, and the headline contrast is +2.25pp with them and without them. The supplementary material (§S4) reports the pooled analysis for a skeptical reader; its gradients are attenuated and in places reversed, because the registration step’s shape is stamped onto every later outcome.

Marks written in language their industry subsequently moved toward fail the five-year proof more often than marks written in language it was moving away from. Within Nice class and registration year, controlling for whether the filer drafted the mark themselves or hired counsel, and for the length of the description, registrations in the most leading fifth of lead fail 1.4 points more often (53.3%) than those in the most lagging fifth (51.9%; $t = 8.3$, Table 3). The primary estimates come from a linear probability model on 3.10 million unique registrations with lead quintiles cut within Nice class and registration year, class×registration-year fixed effects, controls for counsel of record, intent-to-use basis, log description length, log owner filing volume, owner domicile and foreign filing basis, and standard errors clustered on 1.27 million normalized owners.

Table 3: Failure at the five-year proof of continued use and lead: design-based estimates. LPM of event-dated failure on within-class×registration-year lead quintiles (Q1 omitted), unique serials, registrations 2002–2018. Failure is a dated cancellation for non-use at registration age 4.0–8.5 years, under §8 or, for Madrid §66(a) registrations, §71. All specifications include class×registration-year fixed effects; controls are counsel of record, intent-to-use basis, log goods/services length, log owner filing count, owner domicile (US, CN), and foreign basis (§44(e), §66(a)). Base failure rate 52.3%. The same estimates are the main result of a companion paper on pioneering.

Sample	Specification	Q5–Q1 (pp)	SE	n
All	FE only	+2.29	0.16	3,104,534
All	FE + controls	+1.39	0.17	3,104,534
Counsel-represented	FE + controls	+1.15	0.20	2,414,318
Counsel + US-domiciled	FE + controls	+1.60	0.22	1,888,352
Self-filed	FE + controls	+1.80	0.26	690,216
Excluding Madrid §66(a)	FE + controls	+1.82	0.18	2,924,427
Counsel + US, 2002–07	FE + controls	−0.25	0.33	551,989
Counsel + US, 2008–12	FE + controls	+2.00	0.42	551,924
Counsel + US, 2013–18	FE + controls	+2.65	0.25	784,439
All	continuous, pp per sd of lead	+0.58	0.05	3,104,534
Counsel + US	continuous, pp per sd of lead	+0.63	0.06	1,888,352

Standard errors replace significance stars: with n in the millions every coefficient has $p < 0.01$. Errors are clustered on normalized owner; a single owner files many marks, drafts them in a house style, and abandons them together when the business fails, and the within-owner correlation of outcomes at the proof is 0.38, so the 1.27 million owner clusters, not the filings, carry the information.

The gradient is monotone within cells, and the control set matters in a specific way. Self-filers fail the proof at 69.2% against 47.5% for represented owners, and they write shorter, more class-typical, slightly more leading descriptions; without the counsel control, part of any lead gradient would really be a self-filing gradient. With it, the penalty is of the same order in both strata (+1.8pp self-filed, +1.2pp counselled), and it is present in the cleanest stratum, counsel-represented US-domiciled owners (+1.6pp). About forty percent of the raw pooled contrast is composition rather than the within-cell gradient. The penalty is also not constant over the sample: the earliest third of cohorts carries none of it, which is the era variation the companion paper takes as its subject.

Two mechanical readings are checked and fail. *Timing*: the proof is close to a fixed appointment (97% of in-window cancellations post at registration age 6.5–7.0), and on the 3.15 million registrations whose ninth anniversary is inside the record — where failure is settled and no window is needed — the contrast is +2.71pp against the window’s +2.65. *Disengagement*: if the proof measured an owner who lost interest in paperwork rather than a product that left the market, passage should fall with slow office-action responses; it is flat across response-speed quintiles, and the penalty is present in both speed halves.

5.2.1 The penalty under three theme scorings

The identical registrations were re-scored under 500 global themes (ten times the resolution) and under 50 themes fitted separately per class (nothing shared across classes). The lead penalty is a property of the filing, not of the partition (Figure 2): +2.29pp under the production scoring, +2.61 at 500 themes, +1.94 under per-class themes (unclustered SE 0.09 each), positive under all three at once in 28 of the 44 classes with sufficient volume. The scorings agree only moderately filing by filing (lead correlates at 0.64 and 0.52 between partitions), so the penalty survives substantial disagreement about which filings are leading.

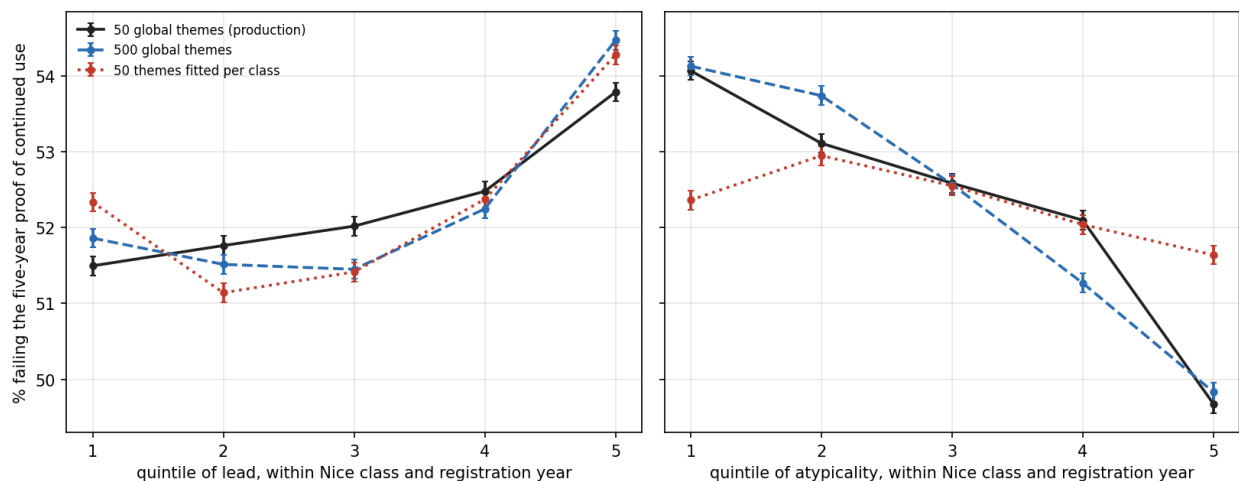


Figure 2: Failure at the five-year proof by quintile of lead (left) and of atypicality (right) under the three theme scorings, on the identical 3.10 million registrations 2002–2018; quintiles cut within Nice class and registration year, no controls. Whiskers are 95% binomial intervals, uncorrected for owner clustering. The lead gradient survives every scoring; the atypicality gradient shrinks sharply under per-class themes.

Atypicality behaves differently. Under the two global scorings the most atypical fifth fails 4.3–4.4 points less often than the least; under per-class themes the gap is 0.7 points. A global model

places every class in one set of coordinates, so a filing that imports a theme the class rarely uses and other classes use often registers as atypical; a model fitted on the class alone re-partitions the class’s own vocabulary, and the imported theme becomes local. The apparent protection of atypicality under global themes is therefore largely a reading of themes imported from other classes. Measuring theme arrival directly, under the 500-theme model on the 4.12 million registration class-records, fits that reading: registrations carrying a theme new to their class (established elsewhere, arriving here) fail the proof 0.43 points *less* within class and year — but 1.06 points *more* once their own lead and atypicality are held. A theme arrives in a class in filings that are atypical for it, and atypical filings survive; among filings of equal atypicality, the ones carrying the arriving theme fail more. Themes new to the whole corpus are too rare to judge (779 registrations). The penalty attaches to being early with language, not to being far from the class’s centre.

Nor is the lead penalty an atypicality effect in disguise. Lead magnitude correlates with atypicality at +0.29, but the signed value does not (−0.05), and cutting lead quintiles within fifths of atypicality leaves the penalty intact in every fifth (+1.7 to +3.3pp).

5.3 The construct is distinct from patenting

Within the firms that do both, trademark vocabulary and patenting are only weakly related. On 94,181 firm-years across 17,506 trademark owners name-matched to PatentsView assignees, regressing each axis on $\log(1 + \text{patents})$ with firms absorbed and errors clustered on the firm puts every coefficient below 0.06 standard deviations. Some are precisely estimated — theme-scored lead is −0.058 ($t = -9.4$) — but all are small, and their signs flip between theme and term scoring. Lagging or leading the patent count by up to two years keeps the coefficients between −0.05 and −0.09, and among owners who both patent and raise a priced round, neither event systematically precedes the other. A coarse pre-assigned sorting of classes into complement and substitute regimes shows no interaction (cell detail in the public repository).

5.4 SEC reporting: the lead profile survives the design; the atypicality gradient is composition

The firm-level outcome is whether an owner ever appears in the Commission’s EDGAR filer index — broader than exchange-listed, and split accordingly below. Raw rates rise steeply across atypicality bins; the gradient does not survive the design (Table 4). Within class and filing year, holding description length, top-quintile atypicality moves SEC reporting by +0.000pp on a 0.481% base. Lead is different: at equal atypicality, being far from the class in either direction helps (+0.067pp per standard deviation of $|L|$, $t = 7.7$), but being on the leading side costs (−0.047pp per standard deviation of L , $t = -7.8$). The raw profile in signed lead is a U whose lagging tail sits higher, which is exactly this pair of effects. Cutting lead fifths within fifths of atypicality, the most lagging fifth out-reports the most leading in four of the five atypicality strata, so the lagging premium is not the mechanical coupling of the two axes. Splitting matched owners on whether their SEC filer number carries a ticker gives 10,556 listed against 9,333 reporting without one, and both halves show the same U, so the result does not depend on counting non-operating registrants as successes. The name-match linkage bounds the atypicality reading in particular: distinctive names match more easily, and name distinctiveness plausibly travels with text distinctiveness.

5.5 Firm-level outcomes depend on the representation

The mark-level results above hold under theme and term scoring alike; the firm-level results do not, and the divergence is the second hazard this paper documents. Under term scoring, leading

Table 4: Debut SEC reporting and vocabulary position: design-based estimates. LPM of P(owner ever in SEC EDGAR) on 1,556,968 registered debut filings 1995–2018, one observation per owner, class×filing-year fixed effects, heteroskedasticity-robust SEs. Quintiles cut within class×year. Base rate 0.481%.

	Coefficient (pp)	SE
<i>Atypicality quintiles, $A = \frac{1}{2}(K^- + K^+)$ (Q1 omitted)</i>		
Q5, FE only	+0.017	0.018
Q5, FE + log length	+0.000	0.018
Q5, FE + log length, on K^- alone	−0.003	0.018
<i>Signed lead quintiles, FE + log length + KL levels</i>		
Q3	−0.133	0.020
Q4	−0.187	0.021
Q5	−0.158	0.024
<i>Decomposition, FE + log length</i>		
lead magnitude $ L $ (pp per sd)	+0.067	0.009
signed lead L (pp per sd)	−0.047	0.006
log description length	+0.056	0.004

filings appear to belong to better firms, consistently across three outcomes: P(EDGAR) rises from 0.29% to 0.42% across term-lead quintiles among registered debuts; trailing term-lead predicts gross margin at +2.2pp per standard deviation on an SEC-matched panel; and firm-year term-lead predicts excess stock returns at one-to-four-year horizons. None of these signed results survives when the same text is scored on themes. (These firm-level comparisons were run on an earlier build of the scores that used calendar-year reference windows. The hazard is a contrast between two representations of the same text on identical samples, so it does not depend on how the windows are drawn.) On identical samples the listing gradient is flat to declining, the margin coefficient reverses sign at every resolution tried, and the past-/future-facing decomposition flips and stays flipped. The term-level listing gradient separately dissolves under document-length controls: within bands of equal length it is U-shaped rather than rising, because listed firms’ counsel write longer descriptions.

One rescue must be ruled out: the theme model’s vocabulary is built from a sample, so rare and emergent terms are invisible to it, and if the firm signal lived exactly there, theme scoring would be erasing a real signal. In the software class, each filing’s invisible-vocabulary share is itself a strong firm marker — but *within* strata of equal invisible share, the signed gradient on the firm outcome is flat to negative, while the mark-level penalty holds in both strata. The firm-level signal that term scoring finds is unsigned specificity: serious firms describe specific products with specific, therefore rare, words, and rare-term mass inflates a word-level divergence mechanically. Raising the theme count does not test this, since a larger T re-partitions the same fixed vocabulary rather than extending it; the invisible-share split is the test that settles the question (the term-versus-theme comparison is developed in Supplementary Material §S2 and §S4). The sign reversal also holds on the current per-filing builds: among the 1.46 million registered debut owners of Table 4’s sample, word-scored lead predicts reaching SEC reporting at +0.035pp per standard deviation ($t = 4.7$) and theme-scored lead at −0.030pp ($t = -4.1$), within class and filing year.

A natural guess is that the specificity is simply professional drafting. It is not, mostly: within class and filing year, counsel of record correlates with the specificity measures only weakly ($r = 0.08$ with log distinct terms, 0.06 with the share of words the theme model does not see, −0.02 with

word-scored lead), and holding counsel leaves the word-scored lead coefficient unchanged. Counsel is, however, a far stronger predictor of the firm outcome than any text measure. Represented debut owners reach SEC reporting at 0.69%, self-filers at 0.15%; within class and filing year, counsel alone explains about fifty times as much of the variation as word-scored lead, and about four and a half times as much as the specificity measures together.

The hazard, stated for reuse: term-resolved text-novelty measures can manufacture firm-performance correlations out of lexical specificity, a distribution-based rescoring — one model fit — is a cheap, decisive diagnostic, and who drafted the filing is a stronger signal of firm quality than anything in its words.

6 Robustness: what changes under other choices

Each result above is stated under one formulation chosen for legibility: fifty global themes, flat weighting inside the reference windows, symmetric five-year windows, estimates conditional on grant, and the 2002–2018 cohorts. Table 5 summarizes what each choice costs; the supplementary material (§S4) carries the full treatment, each variation run on the same filings wherever the variation allows.

Two findings from the sweep bear on any reuse of the measure. First, because a single fit’s lead score carries measurement error of roughly a quarter of its variance at fifty themes (a fifth at 200), every coefficient on lead is attenuated toward zero; correcting for it would enlarge the estimates, not shrink them. Averaging scores over several refits is a cheap way for users of the measure to reduce that noise. Second, lead is a small residual of two large, highly correlated levels ($r = 0.99$), which is what makes it sensitive to anything that inflates the levels — under term scoring, document length — and is why the theme representation, where the level spread is stable, is the one on which a unit of lead means the same thing everywhere.

7 Practices for trademark-text research

Three practices follow for research that reads trademark text. First, score distributions, not words. Word-level divergence is dominated by description length — on this corpus the correlation with log distinct terms is -0.64 — and firm-level performance correlations built on it dissolve or reverse under distribution scoring on identical samples. Any term-level novelty measure should be accompanied by the theme-level rescoring that costs one model fit, and any firm-level analysis should hold counsel of record, which the record observes on every filing and which predicts reaching SEC reporting far more strongly than any text measure. Second, treat the registered description as a conformed document. When an application is abandoned and the applicant tries again, the second attempt is rewritten toward the class’s typical language — from both tails, and whoever drafts it. Samples of registered marks therefore understate the atypicality applicants first attempted, and estimates on registered text are estimates on language that prosecution has already filtered. Studies using description similarity or breadth inherit this compression. Third, the two components of surprise are not interchangeable and should not be pooled. Lead is robust to every partition, weighting, and window tried here; atypicality’s apparent effects can be artifacts of the partition (its protective effect at the five-year proof belongs to theme importation, visible only to global models) or of a control (its listing effect depends on how description length is held). A study that reports “novelty” from trademark text without separating the temporal from the cross-sectional component may attribute to one what belongs to the other.

Table 5: Robustness summary. Baseline: five-year-proof contrast +2.29pp on lead and −4.39pp on atypicality (FE only, quintiles within class×registration year, Table 3). Details and further variations in Supplementary Material §S4.

Variation	Result
Reference weighting (flat vs. 2-year half-life)	Nearly a restatement: lead correlates 0.98 across builds, atypicality 1.000; contrast +2.29 → +1.95pp.
Theme resolution ($T = 50, 100, 200, 500$)	Sign and significance untouched. Corpus-wide the contrast is +2.29, +2.23 and +2.61pp at 50, 200 and 500 themes; in the software class it moves non-monotonically between +2.7 and +5.7pp.
Second random seed (software class)	At 50 themes, per-filing lead reproduces at $r = 0.72$ and the contrast moves +3.23 → +3.89pp; at 200 themes, $r = 0.79$ and +5.20 → +4.88pp. The noise attenuates estimates toward zero.
Representation (themes vs. terms)	Lead contrast +2.29 → +3.56pp; the headline representation is the conservative one. Firm-level results diverge (Section 5.5).
Window width (2–7 years, software class)	Signal rises to five years, flat five to seven; five is defensible rather than optimal.
Asymmetric windows (foresight 3,7 to rupture 7,3)	Penalty rises monotonically +1.65 → +3.00pp; the proof punishes rupture more than foresight, a difference of degree.
Duplicate text (47% of registrations) excluded	Both gradients strengthen: lead +2.29 → +3.05, atypicality −4.39 → −5.68pp. Identical texts cluster at the typical end and dilute; they create nothing.
Without conditioning on grant	The composite is nearly flat because two opposed gradients cancel: registration rises in lead while proof survival falls. Conditioning is what separates them.
Cohort halves	The lead penalty belongs to the 2010–2018 cohorts; 2002–2009 carries a third of it (the era swing, companion paper).
Functional form (AIC over four shapes)	Atypicality profiles are humps at registration and SEC reporting and linear at the proof; lead profiles are close to linear, or V-shaped around a vertex near zero where they bend. The SEC-reporting U in lead decomposes into an atypicality-composition half and a symmetric $ L $ half, with a small negative tilt.
Owner clustering	Within-owner outcome correlation is 0.38 (mechanically 1.0 for listing); regression tables cluster on owner, raw two-sample t -statistics overstate precision roughly twofold.

For users of the classification itself, one caution: internet-bearing filings in goods classes survive at rates closer to the technology classes than to their own, so Nice classes used as industry controls do not partition the internet economy. And for the maintenance data, one asset: the five-year proof of continued use records, for every registered mark, whether its owner was still willing to swear the goods were being sold — a product-level survival outcome with a date on it. The indicator literature has largely left it unused, and it is observable for any cohort more than seven years old at no cost beyond parsing the prosecution history.

8 Access

The code is public at <https://github.com/ericssilver/tm-vocabulary>: the parser from the USPTO case files to dated events, the theme model’s fitting and scoring, the owner links, every analysis script, and every generated figure and table in this paper. The derived tables are published as a versioned data release of the same repository (<https://github.com/ericssilver/tm-vocabulary/releases/tag/data-v1>): the 242 million dated prosecution events and the counsel, filing-basis and declaration fields read alongside them; an event-dated outcome table giving, for every registered mark, the date and registration age of any cancellation at the five-year proof of continued use; the fitted theme model and its vocabulary; the per-filing scores under the production scoring; and the owner links to SEC registrants and Form D issuers. Every table rebuilds from the public USPTO, SEC and PatentsView sources with the scripts provided.

The data can be browsed and explored at <https://aporia.institute/tm-vocabulary/>. Its online appendix, <https://aporia.institute/tm-vocabulary/online-appendix/>, lists every theme with a label, its leading words and its share of each Nice class, gives per-class breakouts of the five-year-proof and registration results, and includes a viewer of each class’s registrations on the lead/atypicality plane. Researchers who want the five-year proof as an outcome without the text measure can take the event-dated outcome table directly; those who want the measure on a different corpus can refit the theme model with the scripts provided.

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Author contributions

Eric Silver: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Competing interests

The author declares no competing interests. The author’s current employment is unrelated to this work, and the views expressed are the author’s alone.

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Data availability

The derived data — the dated prosecution events, the event-dated outcomes, the theme model, and the per-filing scores — are deposited at Zenodo (Silver, 2026). All code and every generated figure and table are public at <https://github.com/ericssilver/tm-vocabulary>, and an interactive per-class online appendix is at <https://aporia.institute/tm-vocabulary/online-appendix/>. The source data are public: the USPTO trademark backfile (United States Patent and Trademark Office, 2026), the SEC’s financial statement data sets, company ticker file and Form D data sets (U.S. Securities and Exchange Commission, 2026a,b,c), and PatentsView (PatentsView, 2024).

Declaration of generative AI in the writing process

This paper was produced in a sustained collaboration with Claude (Anthropic), which drafted and revised text and analysis code under the author’s direction. Every analysis decision, every number against its artifact, and all final text were reviewed by the author, who takes full responsibility for the content.

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Supplementary Material

An Event-Dated Corpus of US Trademark Prosecution and a Two-Sided Measure of Vocabulary Position

Eric Silver

September 2026

This supplement carries the material the article summarizes: registration-step detail (§S1), the term-versus-theme representation comparison (§S2), the burn-in profile (§S3), and the full robustness treatment (§S4). Numbered references to sections, tables and figures without an S prefix point into the article.

S1 Registration detail

S1.1 Registration completion peaks at middling atypicality

Across the scored debut filings (57.6% base rate), completion is an inverse-U in *atypicality*: it climbs from 46.6% at the least atypical end to about 58% in the middle and falls back to 53.4% at the most atypical (Figure S2, panel C). The same shape appears on the past-facing level alone, more mildly, running 54.0% at the bottom quintile to 58.4–59.1% in the middle and 57.7% at the top (Figure S1A).

The signed axis behaves differently. Completion on L falls to a trough of 51.9% just above the median and then rises steadily to 58.0% at the leading end. That trough is composition rather than timing. Lead magnitude correlates with atypicality at +0.31, so the filings near zero lead are disproportionately the low-atypicality filings, and those complete poorly for the reason above. Splitting by atypicality confirms it: the trough is absent in the upper three fifths and survives only in the lowest. In a linear specification the signed axis enters positively — leading filings register slightly more often — but monotonically rather than through an interior peak: what curvature exists at registration belongs to how far a filing’s language sits from its class, not to which side of it the filing sat.

The obvious objection is that unfamiliar language is simply refused more often, so the measure recovers nothing beyond familiarity. Completion falls at *both* ends, so the most class-typical filings also complete less than the middle, which familiarity cannot explain; and the curvature is specific to the unsigned axis, the signed axis showing no interior peak at all.

Refiling after abandonment is itself mildly related to vocabulary position. Refiling rates are flat in atypicality (a middle-minus-tails difference of -0.06pp on a 3.6% base) and peak in the middle of the lead distribution (4.15%), falling to 2.6% among the most lagging abandoned applications and 3.4% among the most leading.

S1.2 Self-filed and professionally filed applications

Counsel of record is observed on every filing. Represented applications are 73% of filings over 1995–2018, falling from 85% in 1995 to 68% in 2018 as online self-filing spread. The two populations file

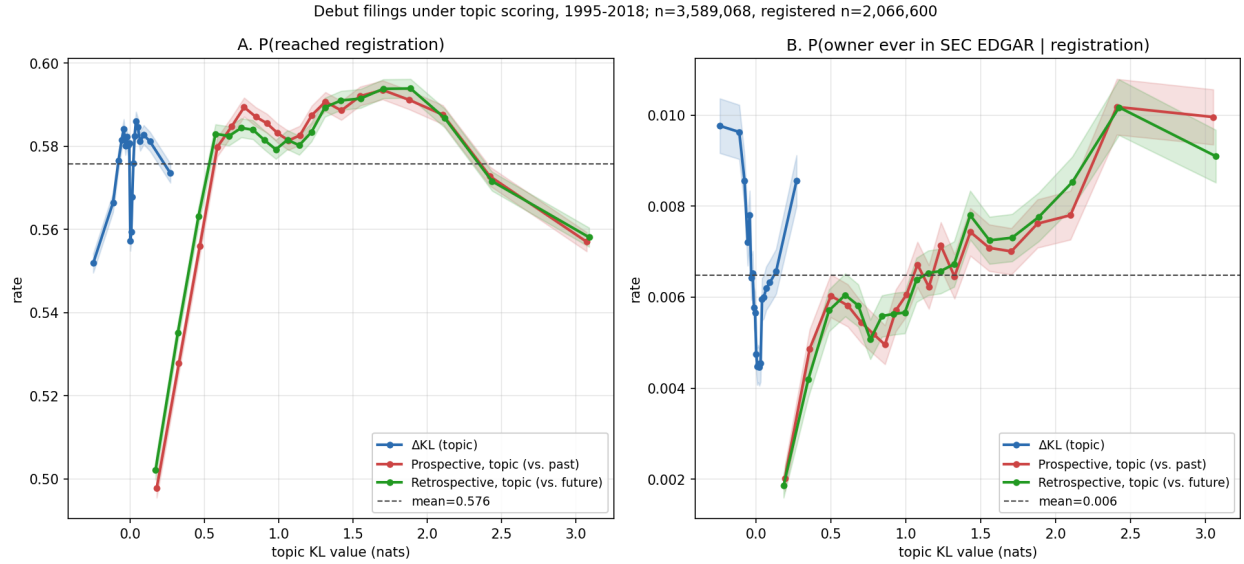


Figure S1: Outcomes for debut filings (owner’s first-ever filing) across all 45 Nice classes under theme scoring, filing years 1995–2018, in quantile bins with 95% intervals, quantiles cut on the pooled distribution. $n = 3,589,068$; registered subset $n = 2,066,600$. *A*: P(reached registration). *B*: P(owner ever in SEC EDGAR, the Commission’s public filing system | registration).

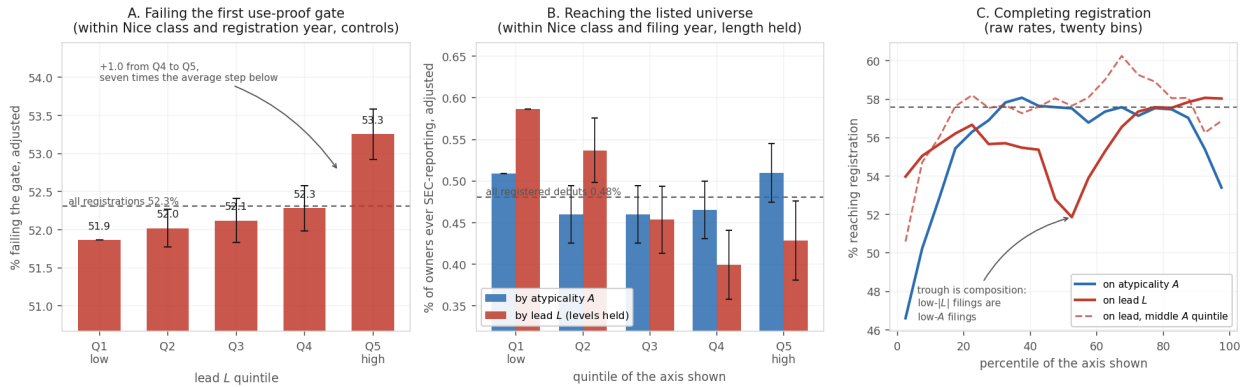


Figure S2: Quintile profiles for the three outcomes. *A*: the share of registrations failing the five-year proof of continued use, by lead quintile, adjusted within Nice class and registration year with the drafting and filer controls of Table 3; the dashed line is the rate for all registrations. The profile rises gently and monotonically through the fourth quintile and then steps up: about seven tenths of the top-to-bottom gap is the single step from the fourth quintile to the fifth, seven times the average step below it. *B*: the share of first-time filers whose owner ever reaches SEC reporting, by quintile of atypicality and of lead, adjusted within Nice class and filing year with description length held. *C*: raw registration rates for debut filings in twenty bins, cut on the pooled distribution — an inverse-U on atypicality, and on lead a trough just above the median that is compositional, absent once atypicality is held in its middle fifth (dashed). Horizontal line at the pooled rate.

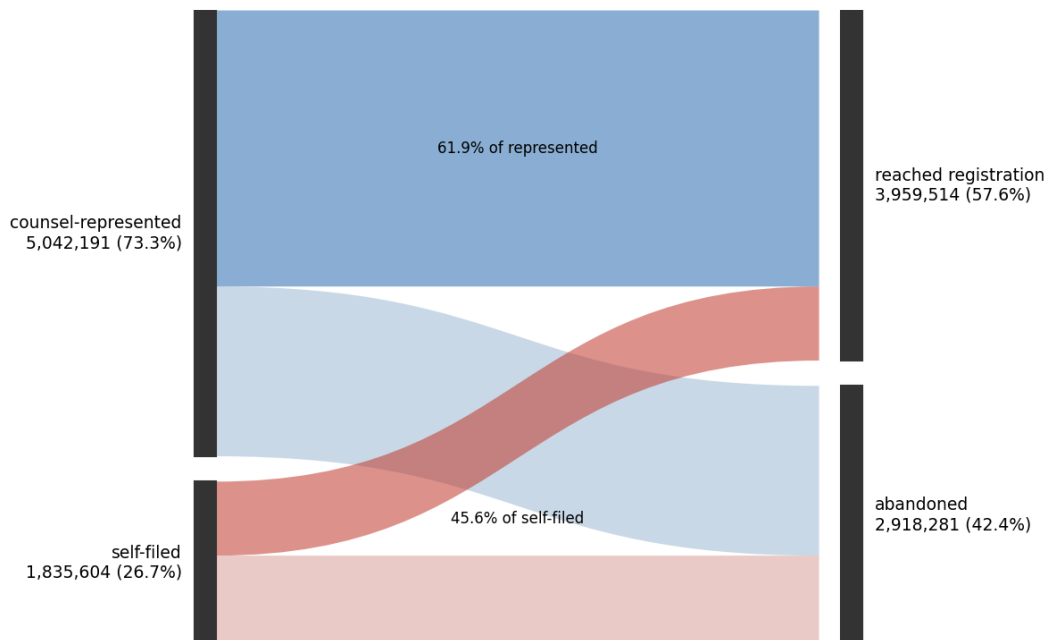


Figure S3: The registration step as a flow, unique serials filed 1995–2018 ($n = 6,877,795$), split by counsel of record on the filing. Represented applications reach registration at 61.9%, self-filed at 45.6%.

differently and fare differently:

	Share of filings	Reached registration	Median terms (009 / 035)
Represented	73.3%	61.9%	28 / 26
Self-filed	26.7%	45.6%	21 / 17

Self-filers write shorter descriptions (a median of 21 terms against 28 in the software class, 17 against 26 in advertising and retail), sit closer to their class’s usual language, and lean slightly further ahead of it. Because who drafts affects both the language and the outcome at once, the five-year-proof regressions control for counsel of record and report their estimates separately for the two groups. Figure S3 draws the full flow from filing to registration split by drafter.

S1.3 Office actions and refiling: applicants rewrite toward the class

An application that is abandoned is usually the end of that attempt: among 2.28 million abandoned applications, 3.6% are followed within six years by the same owner filing the same mark again in the same class, and 1.5% by one that then registers. The refilings that do occur show what an applicant changes. Of the 42,115 abandoned applications whose owner later registered the same mark in the same class, 74% carry a different goods/services description the second time. To see which way the text moved, we score each of these registrations twice: once on the text of the abandoned first attempt, once on the text it registered with. Both self-filers and represented owners rewrite toward the class’s typical description: originals in the most atypical fifth lose 0.39 nats of atypicality, and originals in the most conventional fifth gain 0.16. Refilings that reuse the identical text move by

less than 0.01 at every position, so the convergence is in the writing. Who drafts changes the extent but not the direction: owners who hire counsel for the second attempt rewrite most often (82%) and their descriptions lengthen by 16%; counsel rewriting their own earlier draft rewrite least often (72%) and shorten by 4% (Figure S4). Lead barely moves once we account for the reference windows themselves shifting with the later filing date.

Figure S4 shows the goods/services description before and after a rewrite, for the 31,084 refilings in Section S1.3 whose text changed. A pair is an abandoned application and the same owner’s later registration of the identical mark in the same Nice class within six years; the earliest registered refiling is kept. Counsel of record is read from the prosecution history of each filing separately, so a pair can change representation between attempts. Atypicality is on the production scoring; length is the word count of the normalized description.

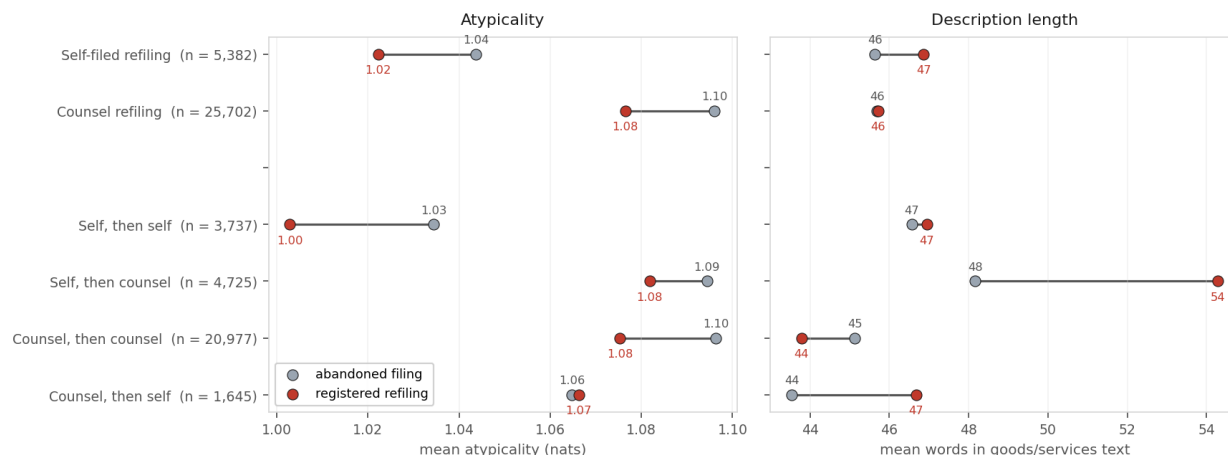


Figure S4: Rewritten refilings: mean atypicality and mean description length of the abandoned filing (grey) and of the registered refiling (red), by representation on the refiling and by the change in representation between the two attempts. Mean atypicality falls by 0.01 to 0.03 nats in every group except counsel-then-self, where it is unchanged. Length rises by six words where counsel is hired for the second attempt and falls by one where counsel rewrites counsel.

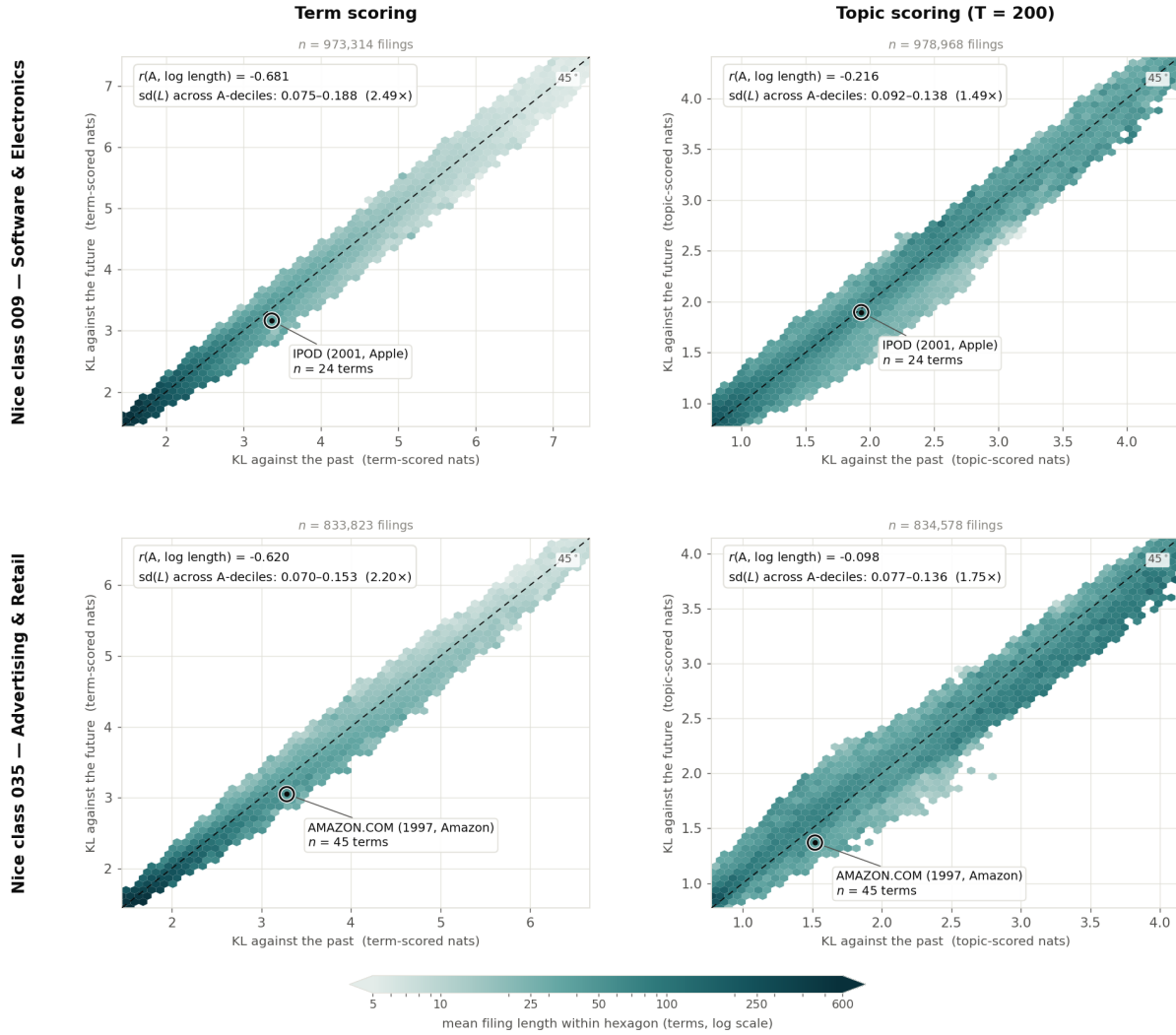
The means understate the convergence, because it runs in both directions. By fifth of the original’s atypicality, the change is +0.16, +0.14, +0.06, −0.06 and −0.39 nats from the most conventional fifth to the most atypical, and within a point of that profile in each of the four representation groups. Identical-text refilings change by +0.002 to +0.008 across the same fifths. On lead, the changed-text profile runs +0.046 to −0.072 and the identical-text profile +0.014 to −0.054; the difference is the rewriting, the rest is the reference windows moving with the later filing date.

S2 Representation: what terms resolve and themes do not

S2.1 Term scoring measures length; theme scoring does not

Figure S5 puts the two representations side by side over the same axes, with each hexagon coloured by the mean length of the filings in it. Under term scoring the colour runs from dark near the origin — a median of about 150 to 170 terms — to near-white at the far end, where the median is eight. That gradient is the measurement problem: a filing scores as atypical largely because it is short. Under theme scoring the same surface is a narrow band of roughly uniform tone. The theme

panels are drawn from the 200-theme build; at the production 50 themes the same check gives a correlation between theme-scored atypicality and log term count of between -0.10 and $+0.10$, against -0.64 under term scoring (article, Section 3).



Hexagons holding fewer than 25 filings are left blank; the retained hexagons still cover over 99.4% of filings in every panel.
The colour scale is shared by all four panels. The axis ranges are not: each panel is boxed on its own 0.5th–99.5th percentiles, and term-scored and topic-scored KL do not share a scale — the comparison invited across columns is of shape and colour gradient, not of raw magnitude.

Figure S5: Mean filing length across the two axes, by representation and class, for filings 1995–2019 scored on per-filing reference windows (theme panels at $T = 200$). The horizontal axis is surprise against the preceding five years, the vertical axis surprise against the following five; a filing on the dashed diagonal is equally far from both, and distance below it is lead. Hexagons holding fewer than 25 filings are blank; the retained hexagons cover over 99.4% of filings in every panel. The colour scale is shared across panels; the axis ranges are not, because term-scored and theme-scored KL are not on a common scale, so only shape and colour gradient are comparable across columns. Both columns are drawn on the filings the per-filing windows score, which differ between representations by under one percent.

Each representation has the virtue the other lacks: theme scoring is independent of document

length, which term scoring is not, while words — the exhibit below counts word pairs as well as single words — show specific new wording, which theme scoring does not. We therefore use theme scoring for every estimate, because the length confound contaminates firm-level comparisons, and words for the worked examples, because the theme basis has no coordinate for the language the examples turn on.

S2.2 How three filings encode

Table S1 traces three filings through term scoring and theme scoring at $T \in \{50, 200, 500\}$ — three kinds of novelty, and how each representation handles them.

AMAZON.COM (1997, advertising/retail) is *combinatorial* novelty: every word is ordinary, but the bigrams assemble an invention (“computerized line,” “line search,” “search ordering”; class document frequencies 379, 134, 355). The filing’s text reads “on line” as two words, the standard spelling in 1997, before *online* lexicalized into a single token; the analyzer drops the stopword “on” and the bigram “line search” survives stopword removal. The novelty was necessarily combinatorial because the word for it did not yet exist. Term scoring sees these bigrams; under theme scoring at every resolution all of them are out of vocabulary and the filing reads as a media retailer that uses computers.

KOMBUCHA (1997, beer/soft drinks) is *lexical* novelty: the new thing arrives as new words (*kombucha* df 567, *saccharose* out of vocabulary even at the class level). The word *kombucha* is in the theme vocabulary but, with no kombucha-like theme, the filing loads on a generic beverages theme at every T (Table S1).

ETHEREUM (2018, software) is *wave* novelty: emergent vocabulary already diffusing (*blockchains*, *distributed computing*). It loads on generic software and services themes at every resolution, including $T = 200$, which does contain a cryptocurrency/blockchain theme; there, the eleven occurrences of *software* dominate the theme assignment.

Table S1: Top theme loading for three filings, by representation. Words show the novel content of all three; theme scoring shows none of it at any tested resolution. Loadings are computed with the model’s fitting tokenizer and can differ from the scoring-time loadings quoted in the article by about 0.01.

	AMAZON.COM (1997)	KOMBUCHA (1997)	ETHEREUM (2018)
novel content (term)	line search, search ordering	kombucha, saccha- rose	blockchains, dis- tributed computing
$T = 50$	video, computer, music (0.40)	beverages, fruit, drinks (0.64)	software, online, downloadable (0.33)
$T = 200$	entertainment, video, games (0.29)	beverages, drinks, coffee (0.60)	field, services, devel- opment (0.37)
$T = 500$	computer, software, data (0.22)	beverages, drinks, fruit (0.61)	software, online, computer (0.49)

The pattern across the row is the measurement point in miniature: theme scoring cannot represent combinatorial novelty (Amazon), dilutes lexical novelty (Kombucha), and majority-votes away wave novelty (Ethereum) even when a fitting theme exists. The mark-level outcome results agree under both scorings despite this blindness, while the firm-level results diverge; that combination is what identifies the firm-level results as artifacts of the lexical specificity that word scoring additionally picks up (article, Section 5.5).

S2.3 Arrangement channels: pair lead and recombination

Combinations can be scored in their own right, at two levels (construction detail in the public repository). The first is *pair lead*: whether the class later moved toward the filing’s arrangements of themes. In the software class it is penalised much as lead is (+2.3 points at the five-year proof, $t = 8.4$, while correlating with lead at only -0.18). But it does not generalize. Rebuilt corpus-wide (where the software class’s net contrast is +1.6), it is zero raw and +0.5pp net of lead and atypicality (owner-clustered $t = 4.2$), significantly positive in 13 of 43 classes and significantly negative in 3 — including -2.1 in advertising and retail — so we report it as a class-specific bound rather than a second construct. The second is word-level *recombination*: adjacent word pairs that are new to the class while both words are established there, novelty assembled from familiar components (Fleming, 2001; Uzzi et al., 2013). In the same class it is rewarded: the top fifth fails 2.8 points less with lead and atypicality held ($t = -10.4$), and it is uncorrelated with pair lead ($r = -0.01$). The construct sees neither arrangement channel, and where they act they run in opposite directions, so its blindness to combination is a bounded omission rather than a hidden bias in one direction.

S3 Burn-in

S3.1 Thin references near the start of the record

Burn-in is the span at each end of the corpus where the reference windows cannot be filled, so scores there reflect a thin reference rather than the text being scored.

A filing’s score depends on having enough same-class filings on each side of it. A full five-year window exists on both sides for filings made 1990 through 2020, and the production scoring runs 1995–2019, holding back five years at the start of that span, where references are thinnest, and one at the end; a filing is also scored only when each of its windows holds at least 500 filings. Figure S6 shows how reference density builds up at the start of the record. How far a thin reference distorts the score was profiled on an earlier build of the measure (word-scored, with calendar-year references): a thin past reference inflates past-facing surprise, and across the classes with enough volume to profile, the median distortion was about 2 nats for filings made in 1990 and fell quickly as the window filled. On that build’s own criterion, though, 18 of 44 classes stayed noisy past 1995, into 1996–2004, so estimates that lean on the earliest cohorts — the companion paper’s era comparisons, in particular — are read with that in mind.

S4 Robustness: what changes under other choices

Each result in Section 5 is stated under one formulation chosen for legibility: one theme resolution, flat weighting inside the reference windows, symmetric five-year windows, estimates conditional on grant, and the 2002–2018 registration cohorts. This section reports what each of those choices costs, and in each case states the more complicated finding beside the simpler one the body uses.

S4.1 Representation and weighting

A scoring is fixed by two choices. The *representation* is what the language is turned into: a distribution over 50 fitted themes, or a distribution over the class vocabulary’s terms. The *weighting* is how days inside the reference window count: every day equal, or decaying by half every two years. Every reference is per-filing, anchored on the filing’s own date (Section 3). Four builds exist:

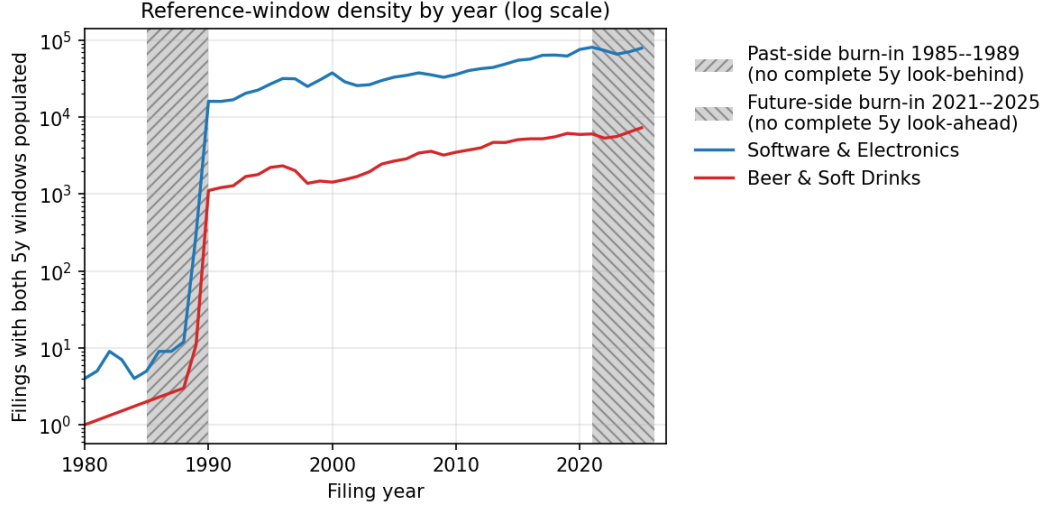


Figure S6: Reference-window density by year on a log scale, for two classes, from the earlier word-scored build. Hatched regions are the past-side and future-side zones where the five-year window is incomplete. The windows decide the score at the filing date; the proof outcome is read separately, at registration age six to seven.

Representation	Weighting	sd(lead)	sd(atypicality)
Themes	flat	0.102	0.698
Themes	half-life 2y	0.082	0.698
Terms	flat	0.155	1.101
Terms	half-life 2y	0.130	1.102

All four are compared on the 6.01 million filings every one of them scored. The first row is the production scoring.

Changing one choice at a time settles the ordering. Correlations of lead between builds differing in exactly one respect:

Choice varied	r (lead)	r (atypicality)
Weighting, flat vs half-life 2y (themes)	0.982	1.000
Weighting, flat vs half-life 2y (terms)	0.970	1.000
Resolution, 50 vs 200 themes ¹	0.636	0.799
Representation, themes vs terms	0.460	0.396

The weighting is close to irrelevant: whether recent language counts for more than distant language within the window barely moves the score, and it does not move atypicality at all to three decimals, which is what the sum-and-difference construction of Section 3 implies — atypicality is a property of the filing, and the weighting acts almost entirely on the small residual that is lead. Resolution and representation matter most: quadrupling the number of themes leaves the two builds agreeing on 40% of the variance in lead, and themes against terms on less than a quarter. A robustness check that varies only the weighting is therefore close to a restatement; one that varies the representation or the resolution is a real test, which is why the term-scored replication in §S4.10 and the sweep below are the ones the claims lean on.

¹The resolution row is computed on the five-year-proof sample rather than the 6.01 million-filing common sample of the other rows.

S4.2 Theme resolution

Fifty themes over a 62,168-term vocabulary is roughly one theme per industry; whether the findings depend on that coarseness is testable by refitting at higher T . Raising the theme count does not add coverage — the vocabulary is fixed, and a larger T re-partitions the same words — but it changes what counts as sitting far from an industry, and could change any sign. Re-fitting the model at four resolutions on the same stratified sample and the same analyzer, and re-scoring the software class under per-filing windows:

Themes	Proof contrast, lead	t	Proof contrast, atypicality	r with $T = 50$
50	+3.23pp	13.8	−14.01pp	—
100	+2.71pp	11.6	−13.69pp	0.690
200	+5.20pp	22.2	−13.45pp	0.640
500	+5.69pp	24.3	−14.49pp	0.641

Three readings, in descending order of how much weight they carry. The sign and the significance are untouched across a tenfold change in resolution, and so is atypicality, which stays within a point of −14pp throughout. The magnitude is not: it does not move monotonically in T , so no single resolution can be quoted as the software class’s effect, and the figure this paper reports at $T = 50$ sits at the low end of the range rather than the high one. (A fifth fit, at $T = 1000$, failed a training-convergence check and is excluded.) And agreement with $T = 50$ settles at 0.64 without decaying as T grows, which says the resolutions share a fixed core and differ in the remainder rather than drifting apart as the partition refines.

The corpus-wide contrast is steadier than the single class: scored for all 45 classes, it is +2.29pp at $T = 50$, +2.23pp at $T = 200$ and +2.61pp at $T = 500$. Pooling evidently averages away much of the per-class resolution sensitivity.

S4.3 Fitting noise at one resolution

The sweep leaves one question open. Scores at different resolutions agree at $r = 0.64$ to 0.69 on lead, and neighbouring resolutions agree no better than distant ones. That has two readings. The number of themes may genuinely change what is being measured. Or a theme-model fit may be unstable, with two runs settling in different local optima, in which case the sweep measures fitting noise and the resolution has little to do with it.

Holding T fixed and moving only the seed separates the two. A second fit at $T = 200$ on the same sample, the same vocabulary and the same number of passes, differing from the first in nothing but its random seed, agrees with it at $r = 0.79$ on lead and $r = 0.87$ on atypicality across 1.11 million scored software-class filings. The cross-resolution figures were 0.64 and 0.83. Most of what the sweep showed as resolution sensitivity is therefore present between two fits that differ only in the seed: for atypicality nearly all of it, for lead well over half. The resolutions are not measuring different things; each fit is a noisy draw on the same thing.

The estimate moves less than the scores do. On the identical sample, the five-year-proof lead contrast is +5.20pp under the first seed and +4.88pp under the second (SE 0.23 each), and the atypicality contrast −13.45 and −14.38pp. Two consequences follow. A single fit’s lead score carries measurement error of roughly a fifth of its variance, which attenuates every coefficient on lead toward zero; correcting for it would enlarge the estimates, not shrink them. And the seed explains most of the disagreement in per-filing scores between resolutions, but not the spread of the class-level estimates: the seed moves the software-class contrast by a third of a point, while

the resolutions in the table above span three. That spread reflects the scorings disagreeing about which filings are leading, and pooling across classes largely averages it out.

The same test at the production resolution gives a split answer. A second fit at $T = 50$, again differing only in its seed, recovers the themes themselves far more faithfully than at $T = 200$: matching each theme to its closest counterpart by the cosine of their word distributions, the median match is 0.74 at fifty themes against 0.54 at 200, and a quarter of the fifty themes match above 0.9. Per-filing scores are no more stable for it: the two fits agree at $r = 0.72$ on lead and 0.84 on atypicality across the same 1.11 million software-class filings, and the five-year-proof lead contrast is +3.23pp under the first seed and +3.89pp under the second (SE 0.23 each). A coarse model’s word lists are reproducible; which theme a borderline filing’s words are credited to is not, and that is where the per-filing noise comes from.

The substantive claim does not depend on any of this. The five-year-proof lead contrast on the 2002–2018 registrations each build scores runs +2.29, +1.95, +3.56 and +2.92pp across the four builds in the order tabulated above, and the atypicality contrast -4.39 , -4.43 , -3.63 and -3.73 pp. Every sign holds; term scoring returns the larger lead contrast, so the headline estimate is the conservative representation.

S4.4 Decayed reference windows

Inside the reference windows every day counts the same. The alternative weights recent language up, with a half-life of two years inside the same 1,826-day truncation, so that a term the class adopted last month is more familiar than one it adopted four years ago. Rescoring the corpus that way changes the scores little and the estimate somewhat. Flat and decayed lead correlate at 0.980 on the 3.10 million five-year-proof registrations and atypicality at 1.000 to three decimals. On the identical registrations, with quintiles cut within class and registration year, the top-minus-bottom proof contrast on lead is +2.29pp under the flat window and +1.95pp under the decayed one (SE 0.09 each); on atypicality it is -4.39 and -4.43 pp. The decayed window therefore returns a lead penalty about a seventh smaller, with the same sign, the same precision, and the same profile. The flat window is kept in the body because it is the simpler statement; a reader who prefers recent language to count for more should take +1.95pp as the estimate.

S4.5 Window width

The window is a parameter of the measure, not a property of it, and the five years used throughout was fixed early in the project rather than tuned to the present construction. To test it, we re-score the software class at several widths under per-filing references and judge each width by the five-year-proof quintile contrast it produces: +2.02pp at two years, +2.20 at three, +3.23 at five and +3.37 at seven, each on an identical sample of 452,911 registrations and each with a standard error of 0.23pp. Signal rises steeply from two to five years and is flat between five and seven: the +1.03pp gain from three to five is well outside the sampling error, the +0.14pp from five to seven is not. A ten-year window returns +2.89pp, but that figure is not comparable to the others — requiring both windows to lie inside the corpus costs 22% of the scored filings and 13% of the proof sample, dropping the earliest and latest registration years, which are also the years whose contrasts differ most. The apparent decline at ten cannot be separated from that change in composition and is not evidence about window width. On the range where the sample is fixed, seven years is nominally the peak and five is within noise of it while retaining more of the corpus, so five is defensible rather than optimal. The sweep is on one class and one outcome.

S4.6 Mixing the two windows

“Leading” is relative to a reference, and the two references need not be the same width. A short past and a long future make lead a measure of foresight — whether the filing resembles where the class was heading over the next seven years more than what it wrote in the last three. A long past and a short future make it a measure of rupture — whether the filing has broken with what the class had established over seven years, judged against only the next three. We score every class on the production representation and per-filing windows at $W \in \{3, 5, 7\}$ years on each side, and evaluate all five variants on identical samples. In the table, the proof columns are computed on the 3.10 million registrations 2002–2018, with quintiles cut within class $\times$ registration-year; the registration columns on the scored filings 1995–2018, with quintiles within class $\times$ filing-year:

Variant	Past, future (years)	Five-year-proof failure		Registration	
		Q5–Q1 (pp)	classes > 0	Q5–Q1 (pp)	Q3 – tails
Foresight	3, 7	+1.65 (0.09)	31 of 44	+1.09	+0.50
Symmetric	3, 3	+1.69 (0.09)	—	+0.79	+0.13
Symmetric	5, 5	+2.29 (0.09)	33 of 44	+0.71	+0.28
Symmetric	7, 7	+2.58 (0.09)	—	+0.48	+0.65
Past-rupture	7, 3	+3.00 (0.09)	37 of 44	+0.01	+0.33

The symmetric five-year row is the paper’s FE-only estimate in Table 3, reproduced here to the decimal, which fixes the scale. The penalty rises monotonically along the mix, from +1.65pp under foresight to +3.00pp under rupture, and rupture is the more pervasive variant, positive in 37 of 44 classes against 31. In software the two are +1.82 and +5.07pp. So the five-year proof punishes distance from the class’s established language more than it punishes resemblance to its coming language; but foresight carries a penalty of its own, more than half the symmetric one, and the separation is a matter of degree rather than kind.

Registration is a different matter. On the signed axis the top-to-bottom contrast is at most a point on a 57% base and the interior-minus-tails depth at most two thirds of a point, under every mix. The examiner’s response to lead, on this scoring, is close to nil.

S4.7 Without conditioning on grant

Figure S7 repeats the lifecycle outcome analysis without the registration conditioning, on filings 2013–2015 pooled across all classes (a window whose registrations land by ~ 2017 and face at least one Section 8 proof by the record’s end in April 2026; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two proofs). Panel A is the unconditional composite, the probability that a filing both registers and passes the proof, measured over all filings including the never-registered. It is nearly flat in lead because two opposed gradients cancel: registration rises in lead (61.3% to 63.2%) while proof survival falls (44.2% to 40.8%); the caption carries the quintile figures. Conditioning on grant is what separates them. Panel B shows the conditional proof outcome on the same filings, and it is U-shaped rather than inverse-U (depth -2.11 pp), with a clearly penalised leading tail consistent with the cleaner single-proof cohort of Figure S8.

S4.8 Toggled filters, windows and parameters

Each analysis in this paper and in the companion paper was rerun under its natural parameter toggles, on the identical data. Nothing reverses a headline claim; three auxiliary readings shift.

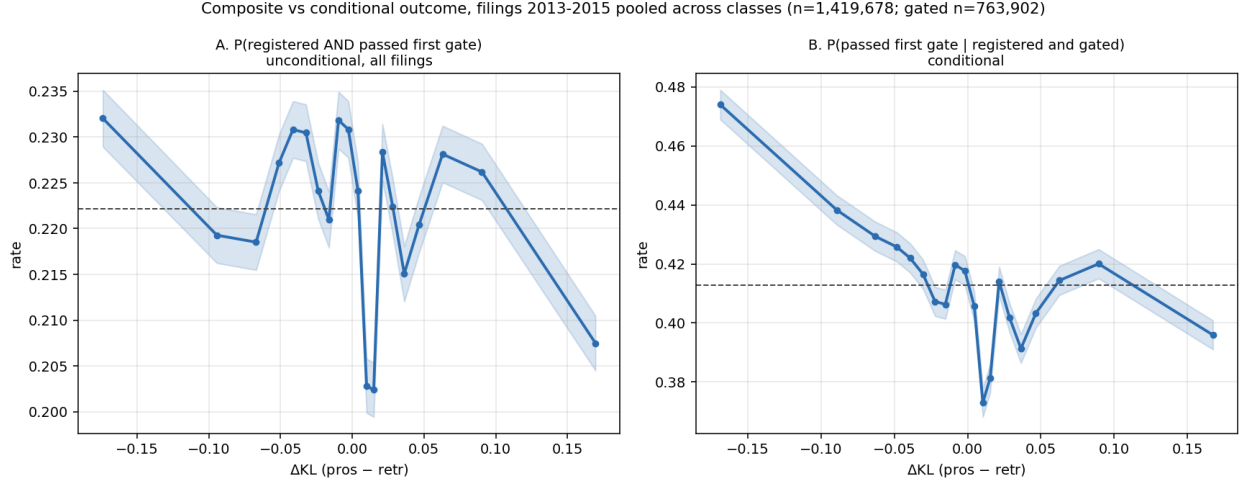


Figure S7: Filings 2013–2015 pooled across classes, theme-scored on per-filing reference windows ($n = 1,419,678$; proof-facing subset $n = 763,902$). *A*: the unconditional composite $P(\text{registered and passed the five-year proof})$ over all filings. *B*: $P(\text{passed} \mid \text{registered and facing the proof})$ on the same filings. The unconditional view is nearly flat in lead (22.4% to 22.1% across quintiles); conditioning on grant is what exposes the gradient.

Variant	Baseline $\rightarrow$ variant	Reading
Gate window 3.5–9.0 / 4.5–8.0	lead +2.29 $\rightarrow$ +2.29 / +2.28pp	window immaterial
Registration cohorts 2002–09	+2.29 $\rightarrow$ +0.98pp	the era swing
Registration cohorts 2010–18	+2.29 $\rightarrow$ +3.11pp	penalty concentrated late
Atypicality, both cohort halves	–4.39 $\rightarrow$ –4.55 / –4.28pp	stable
Internet pattern, narrow core	goods web gap 0.0 $\rightarrow$ +2.0pp	divergence holds
Convergence arrival 1% / 4%	early $\rightarrow$ late +4.4 $\rightarrow$ +1.6	+5.9 $\rightarrow$ +1.8 / +3.8 $\rightarrow$ +2.3
Surge doubling 1.5 $\times$ / 3 $\times$	in-surge net +1.3pp	+1.2 / +0.9pp
Wave entry window 4y / 8y	order flat	flat under both
Recombination floor 2 / 10	net –2.8pp	–1.2 / –3.6pp
Pair-lead presence 0.05	net +2.3pp	+3.1pp
Combination measures, class 035	009 values	recomb –3.8; pair-lead –1.4
Ladder, counsel / self debuts	reporting lead –0.19pp	–0.26 / –0.03pp
Ladder, 2009–13 / 2014–18	reporting atyp. +0.11pp	0.00 / +0.19pp
Excluding duplicate-text filings	lead +2.29; atyp. –4.39pp	+3.05; –5.68pp
Excluding Manual-scale texts only	same	+2.64; –5.05pp

Three of these change what a reader should carry. The lead penalty at the proof belongs to the 2010–2018 cohorts; the 2002–2009 half carries a third of it, which is the era swing the companion paper documents, seen from the other side. The ladder’s gradients belong to counsel-represented debuts — among self-filers the reporting gradient is a null — and its atypicality gradient to the 2014–2018 half. And the pair-lead penalty, +2.3pp in the software class under every parameter tried there, is –2.1pp in advertising and retail on the corpus-wide construction, and pooled across 43 classes is zero raw and +0.5pp net of lead and atypicality, so it is reported as a bound for one class rather than a general fact.

S4.9 Lead is a small residual

Past-facing and future-facing KL are highly correlated however scored. On the complete class-009 corpus under the production scoring they correlate at $r = +0.988$ ($n = 1,109,643$). The lead L is therefore a small residual of two large, stable quantities: both levels have a standard deviation near 0.70 nats about a mean of 1.46, while L has a standard deviation of 0.109 — about a sixth of either level’s spread. Only 1.2% of the two levels’ combined variance survives into their difference. That is what makes L sensitive to anything that inflates the levels; under term scoring the dominant such thing is document length, and Section 3 quantifies the coupling under both representations. The consequence for lead is direct: the standard deviation of L rises 2.2–2.5-fold across deciles of A under term scoring and 1.5–1.8-fold under theme scoring, so a unit of lead means roughly the same thing everywhere only in the theme representation. §S2.1 shows this as a length surface over the two axes.

One qualification: “flat” means no monotone length gradient rather than constant: class 035 under theme scoring is mildly U-shaped in A . The two scorings agree only moderately per filing (Pearson 0.46, Spearman 0.42 on the 6.01 million filings both score). That is what gives their agreement on the mark-level patterns evidential force, and their disagreement on the firm-level patterns diagnostic force (Section 5.5). Across $W \in \{3, 5, 7\}$, 6.9–10.4% of filings flip sign. Per-filing noise attenuates at every aggregation used: firm-year means, token pools, theme prevalence trajectories.

Alignment with expert innovation judgment is at most suggestive. Firms on BCG/MIT “most innovative” lists sit +0.09 standard deviations above the panel mean under term scoring ($p = 0.22$), +0.14 under theme scoring ($p = 0.07$), and +0.08 at $T = 200$ ($p = 0.20$). The matched sample is 41 firms, which is too small to carry weight in either direction, and the paper draws no inference from it. Patent counts remain uncorrelated with lead on the same panel (Pearson +0.010).

S4.10 Term scoring as a cross-check

The snapshot cross-check on the single one-proof-elapsed cohort (registrations 2016–2018, terminal status 710 versus 701–705/800) agrees with the event-dated estimates and is scoring-robust: -5.4 pp survival across quintiles under theme scoring and -5.7 pp under term scoring on the identical sample (Figure S8). Under term scoring, proof survival *rises* in the raw KL level ($+6.5$ pp past-facing, $+7.6$ pp future-facing): atypical text survives the five-year proof better while leading text survives it worse — the same two-axis separation as Section 5.4, at the mark level.

S4.11 Functional form

Four shapes were fitted to each profile by weighted least squares on 40 equal-count bins (Figure S9): linear, quadratic, an exponential $a + be^{cx}$, and a free-vertex symmetric form $a + be^{c|x-x_0|}$ meant to catch a U directly. Selection is by AIC.

Outcome	Axis	R^2 line	R^2 curve	Δ AIC	Shape selected
Registration	atypicality	0.38	0.92	77.1	curve, peak at +1.04
Registration	lead	0.08	0.17	1.9	(quadratic; weak)
Use-proof	atypicality	0.74	0.74	0.0	linear
Use-proof	lead	0.44	0.55	4.3	V, vertex -1.50
SEC reporting	atypicality	0.74	0.84	15.2	curve, peak at +2.17
SEC reporting	lead	0.13	0.59	26.3	V, vertex +0.30
listed only	lead	0.19	0.46	12.2	V, vertex +0.41
reporting, not listed	lead	0.07	0.67	37.2	V, vertex +0.24

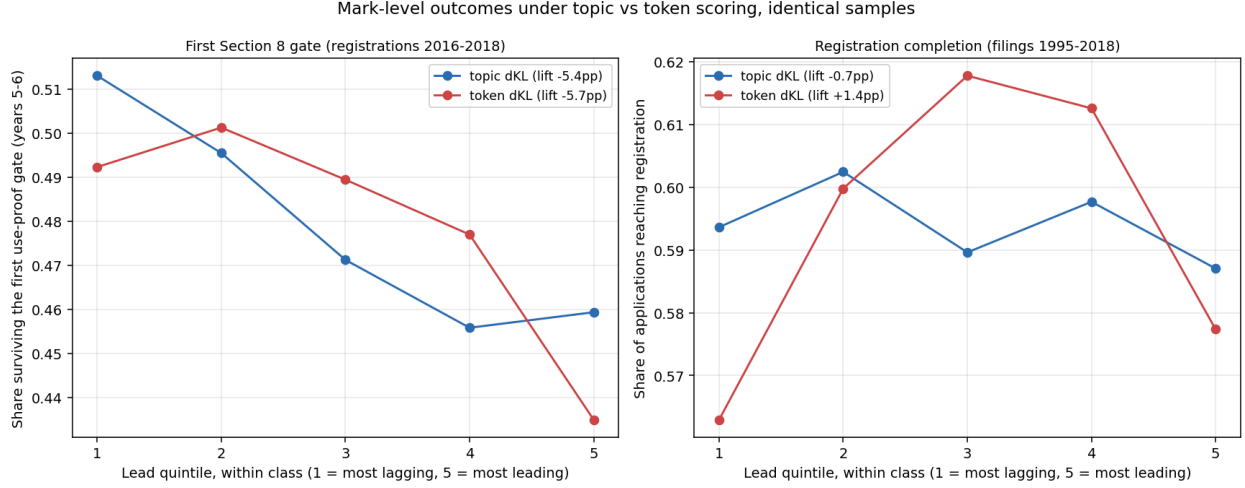


Figure S8: The two mark-level outcomes under theme and term scoring on identical samples, pooled across classes, by within-class lead quintile (1 most lagging, 5 most leading). *Left*: the share of registrations 2016–2018 that survived the five-year proof of continued use at years five to six, read from terminal status codes, where the two representations agree closely. *Right*: registration completion (filings 1995–2018), where they do not: the inverse-U in the signed axis is present under term scoring (depth +0.048) and absent under theme scoring (depth –0.001). The proof result is scoring-robust; the registration curvature on this axis is not, and the registration analysis reports only the unsigned version.

The winning form is a free-vertex $a + be^{c|x-x_0|}$. On the atypicality rows it is a genuine curve, with fitted exponents between 0.5 and 1.2 — a hump that peaks about one standard deviation above mean atypicality at registration and about two above it at SEC reporting. On every lead row the exponent collapses to the order of 10^{-4} , which turns $e^{c|x-x_0|}$ into $1 + c|x - x_0|$: what wins there is a V , linear in distance from a vertex. A V beating a parabola says the turn is sharper than a quadratic allows, not that anything grows exponentially.

Vertices are in standard deviations of the axis across filings.

The two rows for SEC reporting are the outcome Section 5.4 and Table 4 actually run on; the listed and reporting-only splits below them show the same shape on each half, which is why the split does not change the reading.

AIC here is computed on 40 binned means, not on the millions of underlying filings, so it ranks shapes rather than testing them. The 77 at registration and the 12 to 36 at the public-reporting steps are decisive; the 1.9 at registration on lead and the 4.3 at the five-year proof are weak, and those two rows are better read as “no worse than linear” than as curvature.

The non-monotone profiles on signed lead decompose into parts of different symmetry (Figure S10). Each ventile’s deviation of the listing rate from the base is split, by sequential counterfactual prediction, into the part atypicality composition alone predicts, the part exact-duplicate text adds, a linear lead tilt, and a residual. The symmetric U is carried about half by atypicality composition — the spread of L rises with A , so both tails of the lead axis hold atypical filings and the middle holds typical ones — and about half by a residual that is itself symmetric in $|L|$: the same conditional lead magnitude effect Table 4 estimates at +0.067pp per standard deviation. Duplicate-text composition adds almost nothing once atypicality is held, and the linear tilt is an order of magnitude smaller than either symmetric part (–0.07pp per nat). The fitted vertices above follow from this arithmetic: a symmetric profile plus a negative tilt has its minimum shifted right

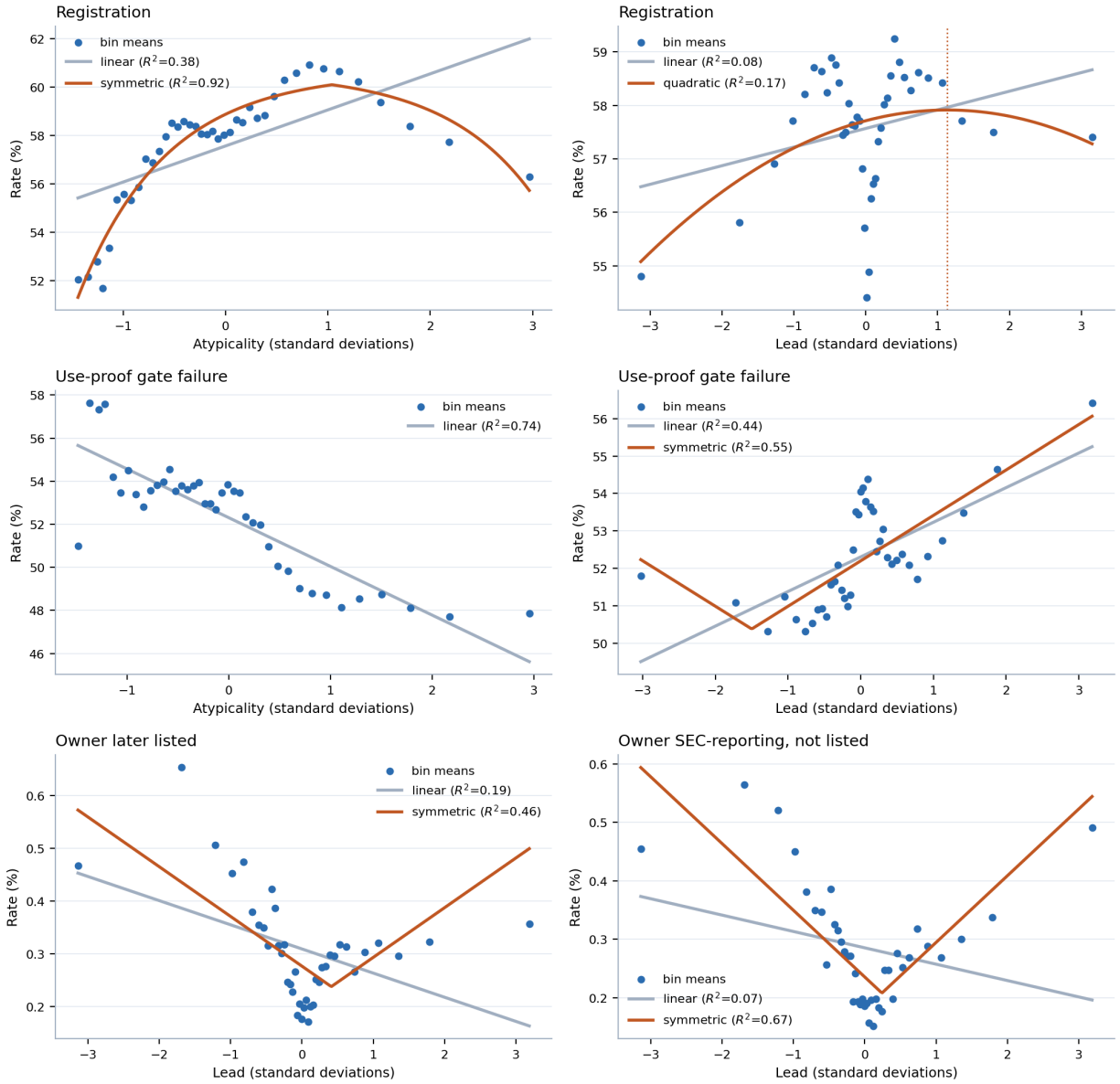


Figure S9: Binned profiles at the three steps, with the linear reading the paper's contrasts assume and the shape selected by AIC where that is not linear. Forty equal-count bins; the dotted line marks the fitted vertex. The bottom row is the case that matters: a linear fit through the public-reporting profile slopes downward while the data turn up at both tails.

of zero, and a survival hump its maximum shifted left, which is where the fits put them (+0.24 to +0.41 standard deviations for the reporting minima; the survival vertex, at -1.5 , comes from a weak fit, $\Delta\text{AIC } 4.3$).

Excluding text-duplicate filings strengthens rather than weakens every headline gradient. Of the five-year-proof sample, 47% of registrations share their exact normalized description with at least one other filing; dropping them raises the lead contrast from +2.29 to +3.05pp and the atypicality contrast from -4.39 to -5.68 pp, and dropping only the Manual-scale texts (clusters of 100 or more identical filings) gives +2.64 and -5.05 pp. Identical texts carry identical scores and cluster at the typical end of the axis, so they dilute both gradients; neither is created by them.

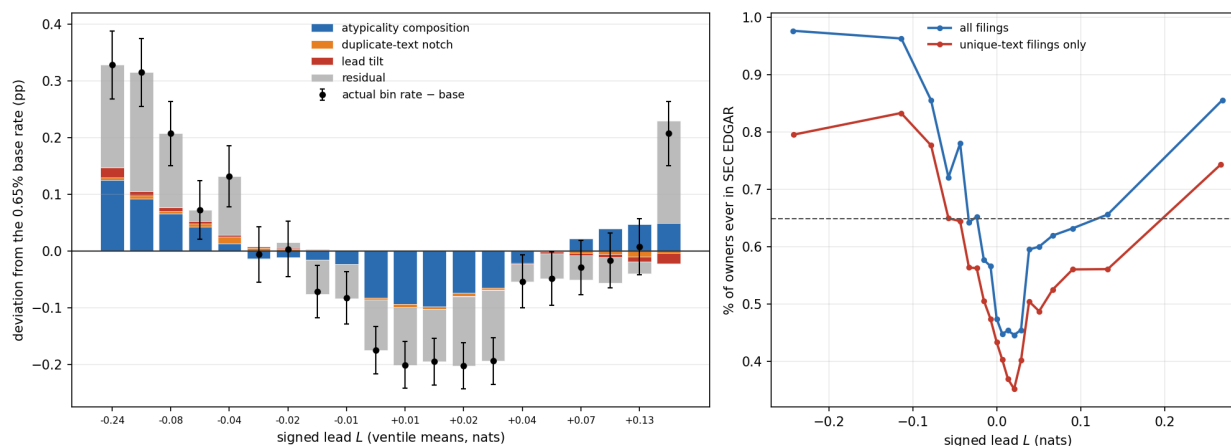


Figure S10: The signed-lead listing profile taken apart. *Left*: each ventile’s deviation of $P(\text{owner ever in SEC EDGAR})$ from the 0.65% base, on 2.07 million registered debuts 1995–2018, split into the part predicted by atypicality composition alone (50 quantile bins of A), the part added by an exact-duplicate-text indicator, a linear tilt in L fitted to the remainder, and the residual; bars stack from zero, dots are the actual deviations with 95% intervals. *Right*: the same profile on all filings and on filings whose normalized description text is unique in their class; the central dip survives, and deepens, without duplicated text.

S4.12 Listing against reporting

“Appears in SEC records” mixes two populations: companies whose shares trade, and entities that file with the Commission without ever listing — private issuers, funds, subsidiaries registering securities. Splitting the 19,889 matched owner strings on whether their CIK carries a ticker in the Commission’s own company-ticker file gives 10,556 listed against 9,333 that file financial statements without one.

It does not change the finding. Among registered debut owners whose debut filing is scored, 0.310% later appear as listed companies and 0.285% as reporting non-listed ones, and both profiles are U-shaped in lead with vertices at +0.41 and +0.24 standard deviations. The listed profile is slightly flatter and noisier, which is what the smaller and more selected outcome predicts. Whatever the signed axis is picking up at this margin, it is not an artifact of counting non-operating registrants as commercial successes, and it is not specific to the act of listing either.²

²Two limits on the split. The ticker file is a current snapshot, so a company that listed in 2003 and was acquired in 2011 appears in the non-listed tier; the tiers are therefore “currently listed” rather than “ever listed,” which will misclassify older successes downward. And the base rates here are computed on the appendix’s own panel, which

S4.13 Filings are not independent draws

Two dependences run through the corpus. Filings can share an exact position, because identical text carries identical scores. And outcomes cluster within owner: the within-owner correlation is 0.38 for proof survival pooled, and it is mechanically 1.0 for public listing, since listing is a property of the owner and a firm with hundreds of marks contributes hundreds of identical successes. The regression tables are unaffected, because the proof regressions cluster errors on normalized owner and the firm-level specifications carry one observation per owner. The raw quintile contrasts of the companion paper — the era, theme, internet and surge tables — are not: their two-sample standard errors ignore the owner clustering, and their t -statistics overstate precision by roughly a factor of two, which the text of those sections carries where the margins are close. Binning filings without the correction likewise overstates how much structure the corpus contains. On a positional grid, correcting cell errors for the clustering design effect — the factor by which within-owner correlation inflates the variance of a cell mean — cuts apparent excess dispersion by roughly a quarter on the proof outcome and by half on listing. Displays of the vocabulary plane are descriptive here for that reason; and since the near-zero pooled correlation between A and L reported in Section 3 does not survive cell by cell within class, per-cell contrasts should not be read as within-class effects.

References

- Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science*, 47(1), 117–132.
- Uzzi, B., Mukherjee, S., Stringer, M., & Jones, B. (2013). Atypical combinations and scientific impact. *Science*, 342(6157), 468–472.

conditions on the debut filing carrying a score and so is smaller than the body's sample.